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Time series 1

Syllabus objectives

- 2.1 Concepts underlying time series models
 - 2.1.1 Explain the concept and general properties of stationary, $I(0)$, and integrated, $I(1)$, univariate time series.
 - 2.1.2 Explain the concept of a stationary random series.
 - 2.1.4 Know the notation for backwards shift operator, backwards difference operator, and the concept of roots of the characteristic equation of time series.
 - 2.1.5 Explain the concepts and basic properties of autoregressive (AR), moving average (MA), autoregressive moving average (ARMA) and autoregressive integrated moving average (ARIMA) time series.
 - 2.1.6 Explain the concepts and properties of discrete random walks and random walks with normally distributed increments, both with and without drift.
 - 2.1.9 Show that certain univariate time series models have the Markov property and describe how to rearrange a univariate time series model as a multivariate Markov model.
- 2.2 Applications of time series models
 - 2.2.3 Describe simple applications of a time series model, including random walk, autoregressive and cointegrated models as applied to security prices and other economic variables.

0 Introduction

The time series material for Subject CS2 has been split into two parts.

This chapter gives the basic definitions and simple examples which we will outline in more detail below.

The sections correspond roughly to the syllabus objectives but not necessarily in the same order.

Firstly, the idea of a time series is introduced. This is just a sequence of observations that we record at regular time intervals. Financial time series would therefore include the closing price of the FTSE 100 index on successive days, the retail prices index in successive months and so on.

This chapter is mainly concerned with a class of processes that is commonly used to model time series data, the so-called *ARIMA* class.

It also deals with some of the necessary background theory and definitions that are required. In particular, stationary processes play a major role. Indeed, it turns out that we can only model time series data efficiently if that data is a realisation of a stationary process. We therefore spend some time discussing the stationarity of various ARIMA processes, and if they are not stationary, how they can be transformed into stationary ones for the process of modelling. The only technique looked at here for transforming a non-stationary process into a stationary one will be differencing the data.

The next chapter concentrates on some of the practical issues surrounding time series analysis:

- is the data set a realisation of a stationary process, and if not, how can the data be transformed?
- fitting a model to the (transformed) data
- forecasting future values.

[Chapter 14](#) also introduces multivariate time series.

1 Properties of a univariate time series

A *univariate time series* is a sequence of observations of a single process taken at a sequence of different times. Such a series can in general be written as:

$$x(t_1), x(t_2), \dots, x(t_n)$$

ie as:

$$\{x(t_i) : i = 1, 2, 3, \dots, n\}$$

Most applications involve observations taken at equally-spaced times. In this case the series is written as:

$$x_1, x_2, \dots, x_n$$

ie as:

$$\{x_t : t = 1, 2, 3, \dots, n\}$$

For instance, a sequence of daily closing prices of a given share constitutes a time series, as does a sequence of monthly inflation figures.

The fact that the observations occur in time order is of prime importance in any attempt to describe, analyse and model time series data. The observations are related to one another and cannot be regarded as observations of independent random variables. It is this very dependence amongst the members of the underlying sequence of variables which any analysis must recognise and exploit.

For example, a list of returns of the stocks in the FTSE 100 index on a particular day is not a time series, and the order of records in the list is irrelevant. At the same time, a list of values of the FTSE 100 index taken at one-minute intervals on a particular day is a time series, and the order of records in the list is of paramount importance.

Note that the observations x_t can arise in different situations. For example:

- the time scale may be inherently discrete (as in the case of a series of 'closing' share prices)
- the series may arise as a sample from a series observable continuously through time (as in the case of hourly readings of atmospheric temperature)
- each observation may represent the results of aggregating a quantity over a period of time (as in the case of a company's total premium income on new business each month).

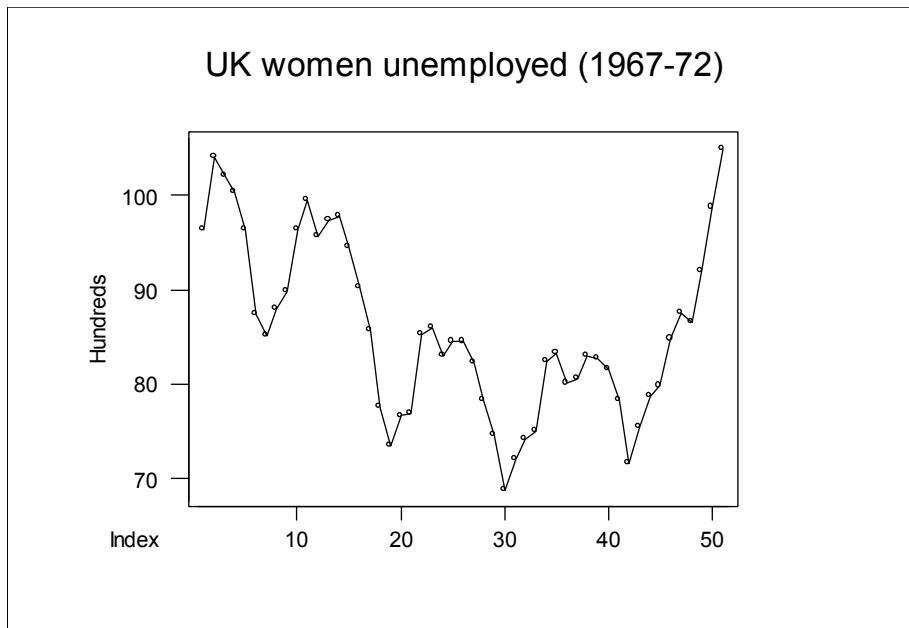


Figure 13.1: a time series

The purposes of a practical time series analysis may be summarised as:

- description of the data
- construction of a model which fits the data
- forecasting future values of the process
- deciding whether the process is out of control, requiring action
- for vector time series, investigating connections between two or more observed processes with the aim of using values of some of the processes to predict those of the others.

These five key aims will be discussed in more detail throughout this chapter.

A univariate time series is modelled as a realisation of a sequence of random variables:

$$\{X_t : t = 1, 2, 3, \dots, n\}$$

called a *time series process*. (Note, however, that in the modern literature the term ‘time series’ is often used to mean both the data and the process of which it is a realisation.) A time series process is a stochastic process indexed in discrete time with a continuous state space.

It is important to appreciate the difference between x_t , which is just a number, and X_t , which is a random variable. The latter will be used to model the former; the procedure for finding a suitable model is the second of our key objectives given above.

The sequence $\{X_t : t = 1, 2, 3, \dots, n\}$ may be regarded as a sub-sequence of a doubly infinite collection $\{X_t : t = \dots, -2, -1, 0, 1, 2, \dots\}$. This interpretation will be found to be helpful in investigating notions such as convergence to equilibrium.

We will often use the shorthand notation X instead of $\{X_t : t = \dots, -2, -1, 0, 1, 2, \dots\}$ or $\{X_t : t = 1, 2, 3, \dots, n\}$.

The phrase, 'convergence to equilibrium' may require some explanation. We will see shortly that a *stationary* process is basically in a (statistical) equilibrium, *ie* the statistical properties of the process remain unchanged as time passes. If a process is currently non-stationary, then it is a natural question to ask whether or not that process will ever settle down and reach (converge to) equilibrium. In this case we can think about what will happen as t gets very large.

Alternatively, we might think of the process as having started some time ago in the past, perhaps indexed by negative t , so that it has already had time to settle down.

This will be made clearer later, when stationarity is discussed in more detail.

2 Stationary random series

2.1 Stationary time series processes

The concept of stationarity was introduced in [Chapter 1](#), along with the ideas of strict and weak stationarity.



Question

Explain what it means for a process X to be:

- (i) strictly stationary
- (ii) weakly stationary.

Solution

- (i) A process X is strictly stationary if the joint distributions of $X_{t_1}, X_{t_2}, \dots, X_{t_n}$ and $X_{k+t_1}, X_{k+t_2}, \dots, X_{k+t_n}$ are identical for all $t_1, t_2, \dots, t_n$ and $k+t_1, k+t_2, \dots, k+t_n$ in J and all integers n . This means that the statistical properties of the process remain unchanged as time elapses.
- (ii) A process X is weakly stationary if $E(X_t)$ is independent of t and $\text{cov}(X_t, X_{t-s})$ depends only on the lag, s .

A weakly stationary process has constant variance since, for such a process, $\text{var}(X_t) = \text{cov}(X_t, X_t)$ is independent of t .

In the study of time series it is a convention that the word ‘stationary’ on its own is a shorthand notation for ‘weakly stationary’, though in the case of a multivariate normal process, strict and weak stationarity are equivalent.

This is because a distribution of a multivariate normal random variable is completely determined by its mean vector and covariance matrix. We will consider this further in [Chapter 14](#).

But we do need to be careful in our definition, as there are some processes which we wish to exclude from consideration but which satisfy the definition of weak stationarity.



Purely indeterministic processes

A process X is called *purely indeterministic* if knowledge of the values of $X_1, \dots, X_n$ is progressively less useful at predicting the value of X_N as $N \rightarrow \infty$. When we talk of a ‘stationary time series process’ we shall mean a weakly stationary purely indeterministic process.



Question

Let Y_t be a sequence of independent standard normal random variables. Determine which of the following processes are stationary time series (given the definition above).

- (i) $X_t = \sin(\omega t + U)$, where U is uniformly distributed on the interval $[0, 2\pi]$
- (ii) $X_t = \sin(\omega t + Y_t)$
- (iii) $X_t = X_{t-1} + Y_t$
- (iv) $X_t = Y_{t-1} + Y_t$
- (v) $X_t = 2 + 3t + 0.5X_{t-1} + Y_t + 0.3Y_{t-1}$

Solution

- (i) For this process, $X_0 = \sin U$, $X_1 = \sin(\omega + U)$, $X_2 = \sin(2\omega + U)$, Given the value of X_0 , future values of the process are fully determined. So this is not purely indeterministic, and is not therefore a stationary time series in the sense defined in the Core Reading.
- (ii) For this process, $X_0 = \sin Y_0$, $X_1 = \sin(\omega + Y_1)$, $X_2 = \sin(2\omega + Y_2)$, Since $E(X_t)$ varies over time, this process is not stationary.

- (iii) Here we have:

$$E(X_t) = E(X_{t-1} + Y_t) = E(X_{t-1}) + E(Y_t) = E(X_{t-1})$$

So the process has a constant mean. However:

$$\text{var}(X_t) = \text{var}(X_{t-1} + Y_t) = \text{var}(X_{t-1}) + \text{var}(Y_t) = \text{var}(X_{t-1}) + 1$$

Here we are using the fact that Y_t is a sequence of independent standard normal random variables. Since the variance is not constant, the process is not stationary.

- (iv) This process is weakly stationary:

$$E(X_t) = E(Y_{t-1}) + E(Y_t) = 0$$

and:

$$\text{cov}(X_t, X_{t+k}) = \begin{cases} 2 & k = 0 \\ 1 & |k| = 1 \\ 0 & |k| \geq 2 \end{cases}$$

For example:

$$\begin{aligned}\text{cov}(X_t, X_t) &= \text{cov}(Y_{t-1} + Y_t, Y_{t-1} + Y_t) \\ &= \text{cov}(Y_{t-1}, Y_{t-1}) + 2\text{cov}(Y_t, Y_{t-1}) + \text{cov}(Y_t, Y_t) \\ &= \text{var}(Y_{t-1}) + \text{var}(Y_t) = 2\end{aligned}$$

In addition, it is purely indeterministic. From the defining equation, we have:

$$X_1 = Y_0 + Y_1$$

So $Y_1 = X_1 - Y_0$ and:

$$X_2 = Y_1 + Y_2 = (X_1 - Y_0) + Y_2$$

Rearranging gives:

$$Y_2 = X_2 - X_1 + Y_0$$

and hence:

$$X_3 = Y_2 + Y_3 = (X_2 - X_1 + Y_0) + Y_3$$

Continuing in this way, we see that:

$$X_n = X_{n-1} - X_{n-2} + X_{n-3} + \cdots + (-1)^{n-2} X_1 + (-1)^{n-1} Y_0 + Y_n$$

From this formula, we see that knowledge of the values of X_1 , X_2 and X_3 , say, becomes progressively less useful in predicting the value of X_n as $n \rightarrow \infty$.

- (v) This process has a deterministic trend via the ' $3t$ ' term, *ie* its mean varies over time. So it is not stationary.

A particular form of notation is used for time series: X is said to be $I(0)$ (read 'integrated of order 0') if it is a stationary time series process, X is $I(1)$ if X itself is not stationary but the increments $Y_t = X_t - X_{t-1}$ form a stationary process, X is $I(2)$ if it is non-stationary but the process Y is $I(1)$, and so on.

We will see plenty of examples of integrated processes when we study the ARIMA class of processes in Section 3.8.

The theory of stationary random processes plays an important role in the theory of time series because the calibration of time series models (that is, estimation of the values of the model's parameters using historical data) can be performed efficiently only in the case of stationary random processes. A non-stationary random process has to be transformed into a stationary one before the calibration can be performed. (See [Chapter 14](#).)



Question

Suppose that we have a sample set of data that looks to be a realisation of an integrated process of order 2. Explain what can we do to the data set in order to model it.

Solution

We can difference the data twice, ie look at the increments of the increments.

2.2 Autocovariance function

The mean function (or trend) of the process is $\mu_t = E[X_t]$, the covariance function $\text{cov}(X_s, X_t) = E[X_s X_t] - E[X_s]E[X_t]$. Both of these functions take a simpler form in the case where X is stationary:

- The mean of a stationary time series process is constant, ie $\mu_t \equiv \mu$ for all t .
- The covariance of any pair of elements X_r and X_s of a stationary sequence X depends only on the difference $r - s$.



Autocovariance function

We can therefore define the autocovariance function $\{\gamma_k : k \in \mathbb{Z}\}$ of a stationary time series process X as follows:

$$\gamma_k \equiv \text{cov}(X_t, X_{t+k}) = E[X_t X_{t+k}] - E[X_t]E[X_{t+k}]$$

The common variance of the elements of a stationary process is given by:

$$\gamma_0 = \text{var}(X_t)$$

If a process is not stationary, then the autocovariance function depends on two variables, namely the time t and the lag k . This could be denoted, for example, $\gamma(t, k) = \text{cov}(X_t, X_{t+k})$. However, one of the main uses of the autocovariance function is to determine the type of process that will be used to model a given set of data. Since this will be done only for stationary series, as mentioned above, it is the autocovariance function for stationary series that is most important.

Because of the importance of the autocovariance function, we will have to calculate it for various processes. This naturally involves calculating covariances and so we need to be familiar with all of the properties of the covariance of two random variables. The following question is included as a revision exercise.



Question

Let X and Y denote any random variables.

(i) Express $\text{cov}(X, Y)$ in terms of $E(X)$, $E(Y)$ and $E(XY)$.

(ii) Express each of the following in terms of $\text{cov}(X, Y)$:

(a) $\text{cov}(Y, X)$

(b) $\text{cov}(X, c)$, where c is a constant

(c) $\text{cov}(2X, 3Y)$

(iii) Give an equivalent expression for $\text{cov}(X, X)$.

(iv) Prove, using your formula in (i), that:

$$\text{cov}(X + Y, W) = \text{cov}(X, W) + \text{cov}(Y, W)$$

(v) Simplify each of the following expressions assuming that $\{X_t\}$ denotes a stationary time series defined at integer times and $\{Z_t\}$ are independent $N(0, \sigma^2)$ random variables.

(a) $\text{cov}(Z_2, Z_3)$

(b) $\text{cov}(Z_3, Z_3)$

(c) $\text{cov}(X_2, Z_3)$

(d) $\text{cov}(X_2, X_3)$

(e) $\text{cov}(X_2, X_2)$

Solution

(i) $\text{cov}(X, Y) = E(XY) - E(X)E(Y)$

(ii) (a) $\text{cov}(Y, X) = \text{cov}(X, Y)$

(b) $\text{cov}(X, c) = 0$

(c) $\text{cov}(2X, 3Y) = 6 \text{cov}(X, Y)$

(iii) $\text{cov}(X, X) = E(X^2) - [E(X)]^2 = \text{var}(X)$

(iv)
$$\begin{aligned} \text{cov}(X + Y, W) &= E[(X + Y)W] - E(X + Y)E(W) \\ &= E(XW + YW) - [E(X) + E(Y)]E(W) \\ &= E(XW) + E(YW) - E(X)E(W) - E(Y)E(W) \\ &= \text{cov}(X, W) + \text{cov}(Y, W) \end{aligned}$$

- (v) (a) $\text{cov}(Z_2, Z_3) = 0$, since they are independent.
- (b) $\text{cov}(Z_3, Z_3) = \text{var}(Z_3) = \sigma^2$
- (c) $\text{cov}(X_2, Z_3) = 0$
- (d) and (e) will depend on the actual process. If it is stationary, then $\text{cov}(X_2, X_3) = \gamma_1$, and $\text{cov}(X_2, X_2) = \gamma_0$.

2.3 Autocorrelation function

The autocovariance function is measured in squared units, so that the values obtained depend on the absolute size of the measurements. We can make this quantity independent of the absolute sizes of X_n by defining a dimensionless quantity, the *autocorrelation function*.



Autocorrelation function

The *autocorrelation function* (ACF) of a stationary process is defined by:

$$\rho_k = \text{corr}(X_t, X_{t+k}) = \frac{\gamma_k}{\gamma_0}$$

The ACF of a purely indeterministic process satisfies $\rho_k \rightarrow 0$ as $k \rightarrow \infty$.

This statement is intuitive. We do not expect two values of a (purely indeterministic) time series to be correlated if they are a long way apart.



Question

Write down the formula for the correlation coefficient between the random variables X and Y .

Hence deduce the formula for the autocorrelation function given above.

Solution

The formula for the correlation coefficient is:

$$\text{corr}(X, Y) = \frac{\text{cov}(X, Y)}{\sqrt{\text{var}(X)\text{var}(Y)}}$$

So:

$$\rho_k = \frac{\text{cov}(X_t, X_{t+k})}{\sqrt{\text{var}(X_t)\text{var}(X_{t+k})}} = \frac{\gamma_k}{\sqrt{\gamma_0 \gamma_0}} = \frac{\gamma_k}{\gamma_0}$$

For a non-stationary process we could define an autocorrelation function by:

$$\rho(t, k) = \frac{\text{cov}(X_t, X_{t+k})}{\sqrt{\text{var}(X_t)}\sqrt{\text{var}(X_{t+k})}} = \frac{\gamma(t, k)}{\sqrt{\gamma(t, 0)}\sqrt{\gamma(t+k, 0)}}$$

However, as with the autocovariance function, it is the stationary case that is of most use in practice.

A simple class of weakly stationary random processes is the white noise processes. A random process $\{e_t : t \in \mathbb{Z}\}$ is a *white noise* if $E[e_t] = 0$ for any t , and:

$$\gamma_k = \text{cov}(e_t, e_{t+k}) = \begin{cases} \sigma^2 & \text{if } k = 0 \\ 0 & \text{otherwise} \end{cases}$$

An important representative of the white noise processes is a sequence of independent normal random variables with common mean 0 and variance σ^2 .

Strictly speaking a white noise process only has to be a sequence of *uncorrelated* random variables, *ie* not necessarily a sequence of *independent* random variables. We can also have white noise processes without zero mean.

Result 13.1

The autocovariance function γ and autocorrelation function ρ of a stationary random process are even functions of k , that is, $\gamma_k = \gamma_{-k}$ and $\rho_k = \rho_{-k}$.

Proof

Since the autocovariance function $\gamma_k = \text{cov}(X_t, X_{t+k})$ does not depend on t , we have:

$$\gamma_k = \text{cov}(X_{t-k}, X_{t-k+k}) = \text{cov}(X_{t-k}, X_t) = \text{cov}(X_t, X_{t-k}) = \gamma_{-k}$$

Thus γ is an even function, which in turn implies that ρ is even.

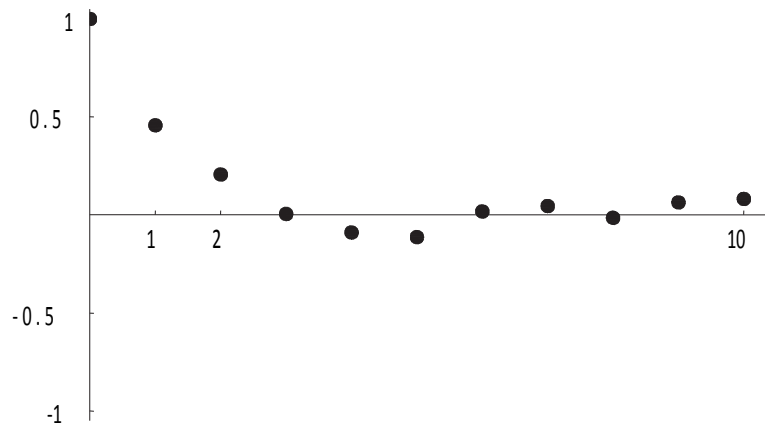
This result allows us to concentrate on positive lags when finding the autocorrelation functions of stationary processes.

2.4 Correlograms

Autocorrelation functions are the most commonly used statistic in time series analysis. A lot of information about a time series can be deduced from a plot of the sample autocorrelation function (as a function of the lag). Such a plot is called a *correlogram*.

Typical stationary series

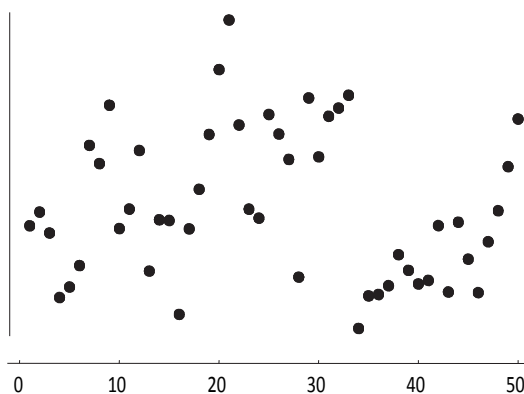
A typical sample autocorrelation function for a stationary series looks like the one shown below. The lag is shown on the horizontal axis, and the autocorrelation on the vertical.



At lag 0 the autocorrelation function takes the value 1, since $\rho_0 = \frac{\gamma_0}{\gamma_0} = 1$. Often the function starts out at 1 but decays fairly quickly, which is indicative of the time series being stationary. The above correlation function tells us that at lags 0, 1 and 2 there is some positive correlation so that a value on one side of the mean will tend to have a couple of values following that are on the same side of the mean. However, beyond lag 2 there is little correlation.

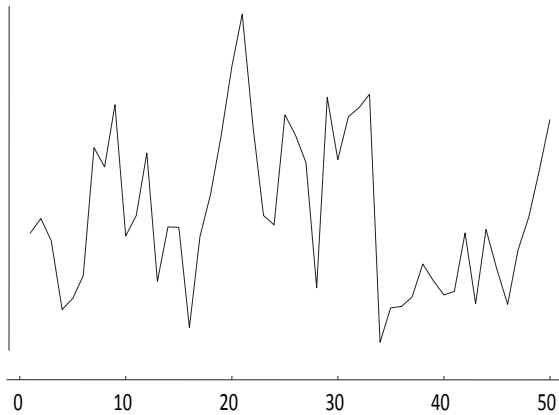
In fact, the above function comes from a sample path of a stationary $AR(1)$ process, namely $X_n = 0.5X_{n-1} + e_n$. (We look in more detail at such processes in the next section.)

The data used for the first 50 values is plotted below. (The actual data used to produce the autocorrelation function used the first 1,000 values.)



The 'gap' in the axes here is deliberate; the vertical axis does not start at zero. The horizontal axis on this and the next graph shows time, and the vertical axis shows the value of the time series X .

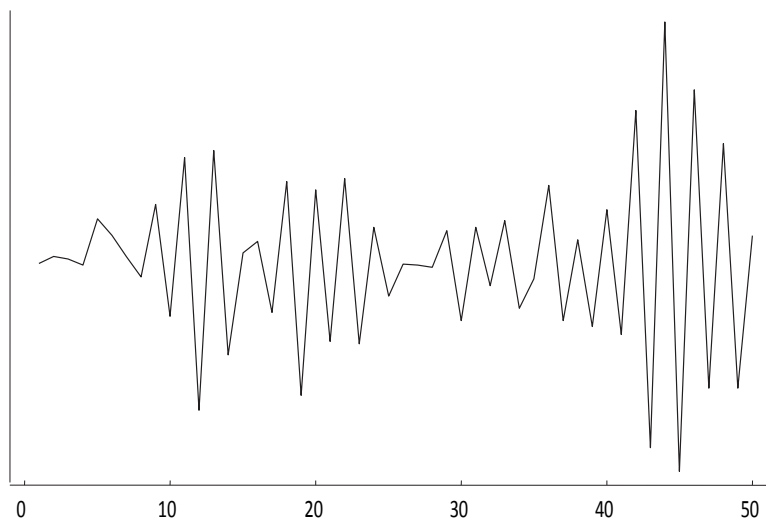
This form of presentation is difficult to interpret. It's easier to see if we 'join the dots'.



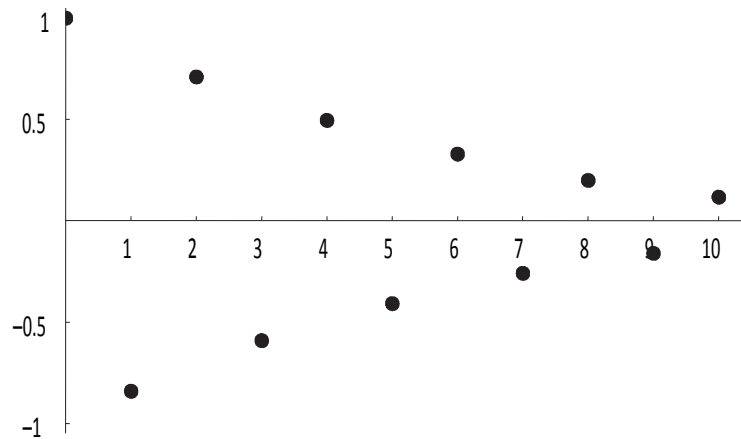
By inspection of this graph we can indeed see that one value tends to be followed by another similar value. This is also true at lag 2, though slightly less clear. Once the lag is 3 or more, there is little correlation.

The previous data set is in stark contrast to the following one.

Alternating series



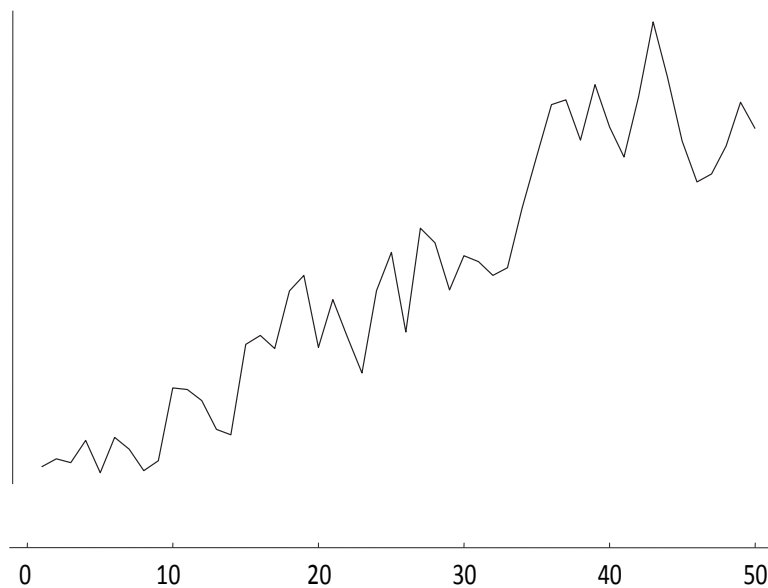
The average of this data is obviously roughly in the middle of the extreme values. Given a particular value, the following one tends to be on the other side of the mean. The series is *alternating*. This is reflected in the autocorrelation function shown below. At lag 1 there is a negative correlation. Conversely, at lag 2, the two points will generally be on the same side of the mean and therefore will have positive correlation, and so on. The autocorrelation therefore also alternates as shown.



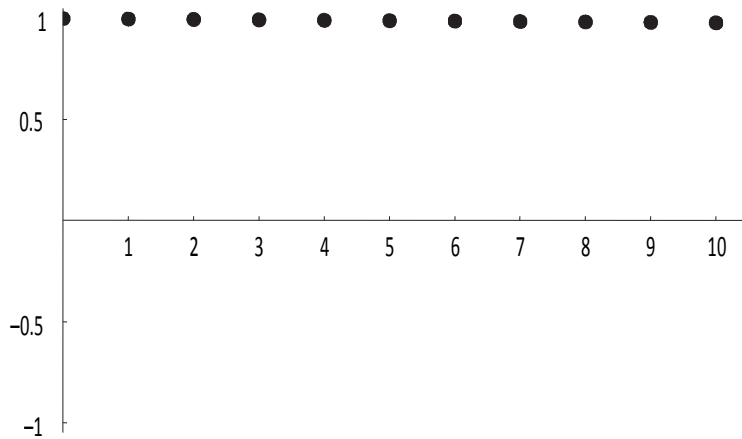
The data in this case actually came from a stationary autoregressive process, this time $X_n = -0.85X_{n-1} + e_n$. This is stationary, but because the coefficient of X_{n-1} is larger in magnitude, ie 0.85 vs 0.5, the decay of the autocorrelation function is slower. This is because the X_{n-1} term is not swamped by the random factors e_n as quickly. It is the fact that the coefficient is negative that makes the series alternate.

Series with a trend

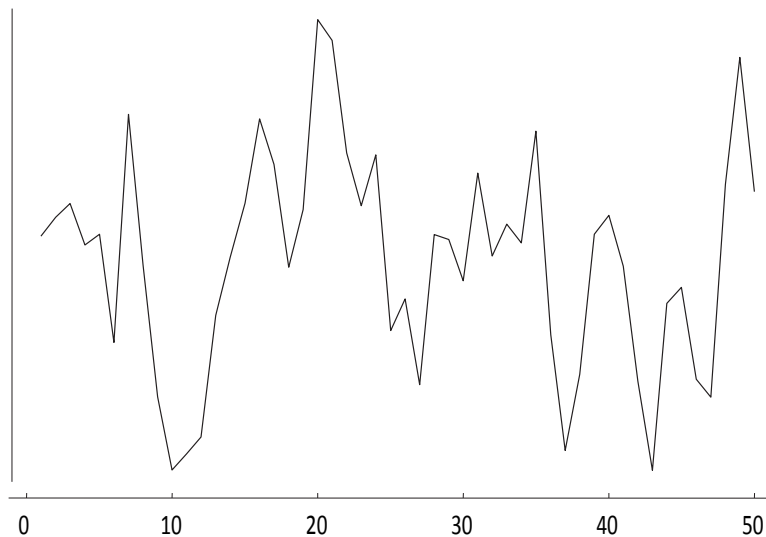
A final example comes from the following data generated from $X_n = 0.1n + 0.5X_{n-1} + e_n$.



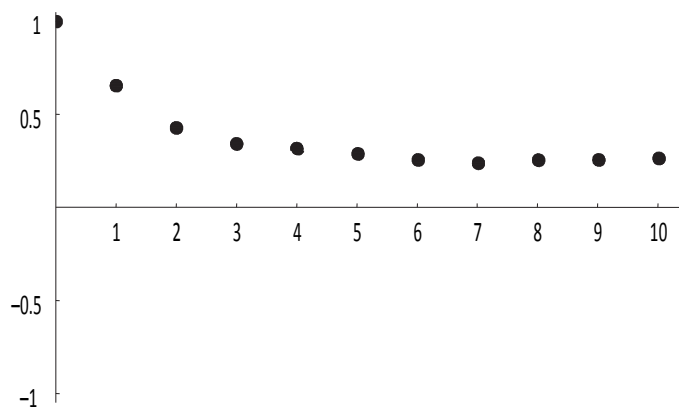
In this time series, a strong trend is clearly visible. The effect of this is that any given value is followed, in general, by terms that are greater. This gives positive correlation at all lags. The decay of the autocorrelation function will be very slow, if it occurs at all.



If the trend is weaker, for example $X_n = 0.001n + 0.5X_{n-1} + e_n$, then there may be some decay at first as the trend is swamped by the other factors, but there will still be some residual correlation at larger lags.



The trend is difficult to see from this small sample of the data but shows up in the autocorrelation function as the residual correlation at higher lags.





Question

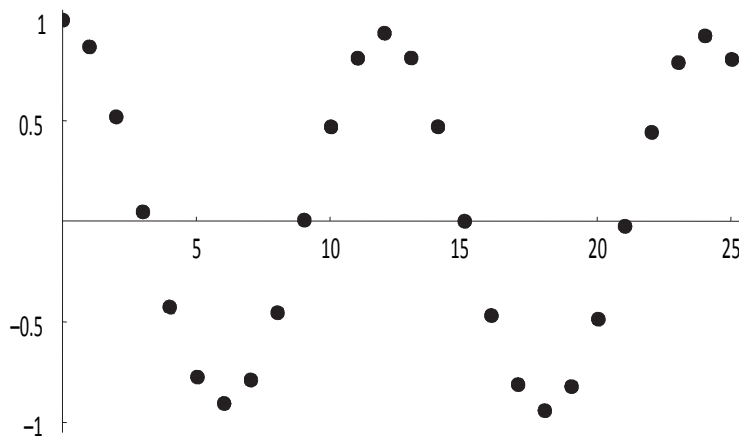
Describe the associations we would expect to find in a time series representing the average daytime temperature in successive months in a particular town, and hence sketch a diagram of the autocorrelation function of this series.

Solution

We expect the temperature in different years to be roughly the same at the same time of year, and hence there should be very strong positive correlation at lags of 12 months, 24 months and so on.

Within each year we would also expect a positive correlation between nearby times, for example with lags of 1 or 2 months, with decreasing correlation as the lag increases. On the other hand, once we reach a lag of 6 months there should be strong *negative* correlation since one temperature will be above the mean, the other below it. For example comparing June with December.

The autocorrelation function will therefore oscillate with period of 12 months.



2.5 Partial autocorrelation function

Another important characteristic of a stationary random process is the *partial autocorrelation function (PACF)*, $\{\phi_k : k = 1, 2, \dots\}$, defined as the conditional correlation of X_{t+k} with X_t given $X_{t+1}, \dots, X_{t+k-1}$.

Unlike the autocovariance and autocorrelation functions, the PACF is defined for positive lags only.

This may be derived as the coefficient $\phi_{k,k}$ in the problem to minimise:

$$E \left[\left(X_t - \phi_{k,1} X_{t-1} - \phi_{k,2} X_{t-2} - \dots - \phi_{k,k} X_{t-k} \right)^2 \right]$$

We can explain the last expression as follows. Suppose that at time $t-1$ we are trying to estimate X_t , but we are going to limit our choice of estimator to linear functions of the k previous values $X_{t-k}, \dots, X_{t-1}$. The most general linear estimator will be of the form:

$$\phi_{k,1}X_{t-1} + \phi_{k,2}X_{t-2} + \dots + \phi_{k,k}X_{t-k}$$

where $\phi_{k,i}$ are constants. We can choose the coefficients to minimise the mean square error, which is the expression given above in Core Reading. The partial autocorrelation for lag k is then the weight that we assign to the X_{t-k} term.



Question

Consider the process $X_t = 0.5X_{t-2} + e_t$, where e_t forms a white noise process.

Determine the partial autocorrelation function for this process.

Solution

For $k=1$ we just have the correlation itself. However, in this case it is clear that the X_t for even values of t are independent of those for odd values. It follows that the correlation at lag 1 is 0.

For $k=2$ the partial autocorrelation is the coefficient of X_{t-2} in the best linear estimator:

$$\phi_{2,1}X_{t-1} + \phi_{2,2}X_{t-2}$$

Comparing this to the defining equation suggests that $\phi_2 = 0.5$.

Similarly, the defining equation suggests that the best linear estimator will not involve $X_{t-3}, X_{t-4}, \dots$. It follows that for $k \geq 3$, we have $\phi_k = 0$.

For the time series in the previous question, we have $\phi_4 = 0$. This is in contrast to the actual correlation at lag four, since X_t depends on X_{t-2} , which in turn depends on X_{t-4} . X_t and X_{t-4} will therefore be correlated. The partial autocorrelation is zero, however, because it effectively removes the impact of the correlation at smaller lags.

In general it is difficult to calculate the PACF by hand.

The formula for calculating ϕ_k involves a ratio of determinants of large matrices whose entries are determined by $\rho_1, \dots, \rho_k$; it may be found in standard works on time series analysis, and is readily available in common computer packages like R.

The diagrams below show the autocorrelation function and partial autocorrelation of an $ARMA(1,1)$ series. ARMA processes are discussed in detail in Section 3.7.

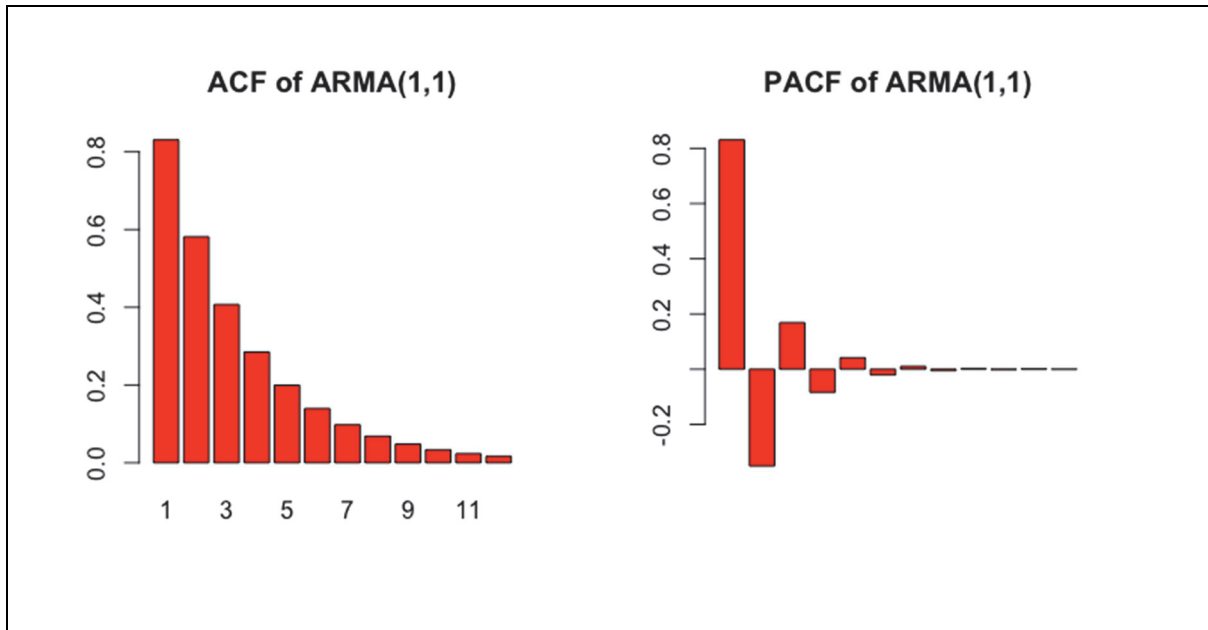


Figure 13.1: ACF and PACF values of some stationary time series model.

In particular the formulae for ϕ_1 and ϕ_2 are as follows:



Partial autocorrelation function at lags 1 and 2

$$\phi_1 = \rho_1, \quad \phi_2 = \frac{\det \begin{pmatrix} 1 & \rho_1 \\ \rho_1 & \rho_2 \end{pmatrix}}{\det \begin{pmatrix} 1 & \rho_1 \\ \rho_1 & 1 \end{pmatrix}} = \frac{\rho_2 - \rho_1^2}{1 - \rho_1^2}$$

Note that for each k , ϕ_k depends on only $\rho_1, \rho_2, \dots, \rho_k$.

These formulae can be found on page 40 of the *Tables*. Their derivations are not required.

It is important to realise that the PACF is determined by the ACF, as the above expressions suggest. The PACF does not therefore contain any extra information; it just gives an alternative presentation of the same information. However, as we will see, this can be used to identify certain types of process.

3 Main linear models of time series

3.1 Introduction

The main linear models used for modelling stationary time series are:

- Autoregressive process (*AR*)
- Moving average process (*MA*)
- Autoregressive moving average process (*ARMA*).

The definitions of each of these processes, presented below, involve the standard zero-mean white noise process $\{e_t : t = 1, 2, \dots\}$ defined in Section 2.3.

In practice we often wish to model processes which are not $I(0)$ (stationary) but $I(1)$. For this purpose a further model is considered:

- Autoregressive integrated moving average (*ARIMA*).

ARMA and *ARIMA* are pronounced as single words (similar to 'armour' and 'areema').

Autoregressive

An *autoregressive process* of order p (the notation $AR(p)$ is commonly used) is a sequence of random variables $\{X_t\}$ defined consecutively by the rule:

$$X_t = \mu + \alpha_1(X_{t-1} - \mu) + \alpha_2(X_{t-2} - \mu) + \dots + \alpha_p(X_{t-p} - \mu) + e_t$$

Thus the autoregressive model attempts to explain the current value of X as a linear combination of past values with some additional externally generated random variation. The similarity to the procedure of linear regression is clear, and explains the origin of the name 'autoregression'.

Moving average

A *moving average process* of order q , denoted $MA(q)$, is a sequence $\{X_t\}$ defined by the rule:

$$X_t = \mu + e_t + \beta_1 e_{t-1} + \dots + \beta_q e_{t-q}$$

The moving average model explains the relationship between the X_t as an indirect effect, arising from the fact that the current value of the process results from the recently passed random error terms as well as the current one. In this sense, X_t is 'smoothed noise'.

Autoregressive moving average

The two basic processes (AR and MA) can be combined to give an autoregressive moving average, or *ARMA*, process. The defining equation of an $ARMA(p, q)$ process is:

$$X_t = \mu + \alpha_1(X_{t-1} - \mu) + \dots + \alpha_p(X_{t-p} - \mu) + e_t + \beta_1 e_{t-1} + \dots + \beta_q e_{t-q}$$

Note: $ARMA(p,0)$ is $AR(p)$; $ARMA(0,q)$ is $MA(q)$.

Autoregressive integrated moving average

The definition of an $ARIMA(p,d,q)$ process is given in Section 3.8.

3.2 The backwards shift operator, B , and the difference operator, ∇

Further discussion of the various models will be helped by the use of two operators which operate on the whole time series process X .

The **backwards shift operator**, B , acts on the process X to give a process BX such that:

$$(BX)_t = X_{t-1}$$

If we apply the backwards shift operator to a constant, then it doesn't change it:

$$B\mu = \mu$$

The **difference operator**, ∇ , is defined as $\nabla = 1 - B$, or in other words:

$$(\nabla X)_t = X_t - X_{t-1}$$

Both operators can be applied repeatedly. For example:

$$(B^2 X)_t = (B(BX))_t = (BX)_{t-1} = X_{t-2}$$

$$(\nabla^2 X)_t = (\nabla X)_t - (\nabla X)_{t-1} = X_t - 2X_{t-1} + X_{t-2}$$

and can be combined as, for example:

$$(B\nabla X)_t = (B(1-B)X)_t = (BX)_t - (B^2 X)_t = X_{t-1} - X_{t-2}$$

The usefulness of both of these operators will become apparent in later sections.

We could also work out $\nabla^2 X_t$ as follows:

$$\nabla^2 X_t = (1-B)^2 X_t = (1-2B+B^2)X_t = X_t - 2X_{t-1} + X_{t-2}$$

Similarly:

$$\nabla^3 X_t = (1-B)^3 X_t = (1-3B+3B^2-B^3)X_t = X_t - 3X_{t-1} + 3X_{t-2} - X_{t-3}$$

In addition, we use the difference operator to write $X_t - 5X_{t-1} + 7X_{t-2} - 3X_{t-3}$ as follows:

$$\begin{aligned} X_t - 5X_{t-1} + 7X_{t-2} - 3X_{t-3} &= (X_t - X_{t-1}) - 4(X_{t-1} - X_{t-2}) + 3(X_{t-2} - X_{t-3}) \\ &= \nabla X_t - 4\nabla X_{t-1} + 3\nabla X_{t-2} \\ &= (\nabla X_t - \nabla X_{t-1}) - 3(\nabla X_{t-1} - \nabla X_{t-2}) \\ &= \nabla^2 X_t - 3\nabla^2 X_{t-1} \end{aligned}$$



Question

Suppose that $w_n = \nabla x_n$. Give a formula for x_n in terms of x_0 and the differences:

$$w_n = x_n - x_{n-1}, \quad w_{n-1} = x_{n-1} - x_{n-2}, \dots, \quad w_1 = x_1 - x_0$$

Solution

We have:

$$\begin{aligned} x_n &= w_n + x_{n-1} \\ &= w_n + w_{n-1} + x_{n-2} \\ &= \dots \\ &= w_n + w_{n-1} + \dots + w_1 + x_0 \\ &= x_0 + \sum_{i=1}^n w_i \end{aligned}$$



The R commands for generating the differenced values of some time series x are:

```
diff(x, lag=1, differences=1)
```

for ordinary difference ∇ .

```
diff(x, lag=1, differences=3)
```

for differencing three times ∇^3 , and:

```
diff(x, lag=12, differences=1)
```

for a simple seasonal difference with period 12, ∇_{12} (see [Section 1.4](#) in Chapter 14).

3.3 The first-order autoregressive model, $AR(1)$

The simplest autoregressive process is the $AR(1)$, given by:

$$X_t = \mu + \alpha(X_{t-1} - \mu) + e_t \quad (13.1)$$

A process satisfying this recursive definition can be represented as:

$$X_t = \mu + \alpha^t(X_0 - \mu) + \sum_{j=0}^{t-1} \alpha^j e_{t-j} \quad (13.2)$$

This representation can be obtained by substituting in for X_{t-1} , then for X_{t-2} , and so on, until we reach X_0 . It is important to realise that X_0 itself will be a random variable in general – it is not necessarily a given constant, although it might be.

It follows that the mean function μ_t is given by:

$$\mu_t = \mu + \alpha^t (\mu_0 - \mu)$$

Here the notation μ_t is being used in place of $E(X_t)$. This result follows by taking expectations of both sides of Equation (13.2), and noting that the white noise terms have zero mean.

The white noise terms are also uncorrelated with each other, and with X_0 . It follows that the variance of X_t can be found by summing the variances of the terms on the right-hand side.

The same representation (13.2) gives the variance:

$$\text{var}(X_t) = \sigma^2 \frac{1 - \alpha^{2t}}{1 - \alpha^2} + \alpha^{2t} \text{var}(X_0)$$

where, as before, σ^2 denotes the common variance of the white noise terms $\{e_t\}$.



Question

Derive this expression.

Solution

From Equation (13.2), we have:

$$\begin{aligned} \text{var}(X_t) &= \text{var} \left(\mu + \alpha^t (X_0 - \mu) + \sum_{j=0}^{t-1} \alpha^j e_{t-j} \right) \\ &= \alpha^{2t} \text{var}(X_0 - \mu) + \sum_{j=0}^{t-1} \alpha^{2j} \text{var}(e_{t-j}) \\ &= \alpha^{2t} \text{var}(X_0) + \sigma^2 \sum_{j=0}^{t-1} \alpha^{2j} \end{aligned}$$

Now using the formula $a + ar + ar^2 + \dots + ar^{n-1} = \frac{a(1-r^n)}{1-r}$ for summing the first n terms of a geometric progression, we see that:

$$\text{var}(X_t) = \alpha^{2t} \text{var}(X_0) + \sigma^2 \left(\frac{1 - \alpha^{2t}}{1 - \alpha^2} \right)$$

For the process X to be stationary, its mean and variance must both be constant. A quick look at the expressions above is enough to see that this will not be the case in general. It is therefore natural to ask for conditions under which the process is stationary.

From this it follows that a stationary process X satisfying (13.1) can exist only if $|\alpha| < 1$.

Further requirements are that $\mu_0 = \mu$ and that $\text{var}(X_0) = \sigma^2 / (1 - \alpha^2)$.

It should be clear that we require $\mu_0 = \mu$ in order to remove the t -dependence from the mean.

Similarly, we require $\text{var}(X_0) = \frac{\sigma^2}{1 - \alpha^2}$ in order to make the variance constant. (We are assuming $\alpha \neq 0$, otherwise X is a white noise process, which is certainly stationary.)

We also require $|\alpha| < 1$. One way of seeing this is to note that the variance has to be a finite non-negative number.

Notice that this implies that X can be stationary only if X_0 is random. If X_0 is a known constant, then $\text{var}(X_0) = 0$ and $\text{var}(X_t)$ is no longer independent of t , whereas if X_0 has expectation different from μ then the process X will have non-constant expectation.

We now consider the situation in which $\mu_t \neq \mu$, and/or $\text{var}(X_0) \neq \frac{\sigma^2}{1 - \alpha^2}$. From what we've just said, the process will then be non-stationary. However, what we are about to see is that even if the process is non-stationary, as long as $|\alpha| < 1$, the process will become stationary in the long run, without any extra conditions.

It is easy to see that the difference $\mu_t - \mu$ is a multiple of α^t and that $\text{var}(X_t) - \sigma^2 / (1 - \alpha^2)$ is a multiple of α^{2t} .

This follows by writing the equations we derived above for the mean and variance in the form:

$$\mu_t - \mu = \alpha^t (\mu_0 - \mu) \quad \text{and} \quad \text{var}(X_t) - \frac{\sigma^2}{1 - \alpha^2} = \alpha^{2t} \left[\text{var}(X_0) - \frac{\sigma^2}{1 - \alpha^2} \right]$$

Both of these terms will decay away to zero for large t if $|\alpha| < 1$, implying that X will be virtually stationary for large t .

We can also turn this result on its head: if we assume that the process has already been running for a very long time, then the process will be stationary. In other words, any $AR(1)$ process with an infinite history and $|\alpha| < 1$ will be stationary.

In this context it is often helpful to assume that $X_1, \dots, X_n$ is merely a sub-sequence of a process $\dots, X_{-1}, X_0, X_1, \dots, X_n$ which has been going on unobserved for a long time and has already reached a 'steady state' by the time of the first observation. A double-sided infinite process satisfying (13.1) can be represented as:

$$X_t = \mu + \sum_{j=0}^{\infty} \alpha^j e_{t-j} \tag{13.3}$$

This is an explicit representation of the value of X_t in terms of the historic values of the process e . The infinite sum on the right-hand side only makes sense, *ie* converges, if $|\alpha| < 1$. The equation can be derived in two ways – either using an iterative procedure as used above to derive Equation (13.2), or by using the backward shift operator.

For the latter method, we write the defining equation $X_t = \mu + \alpha(X_{t-1} - \mu) + e_t$ in the form:

$$(1 - \alpha B)(X_t - \mu) = e_t$$

The expression $(1 - \alpha B)$ will be invertible (using an expansion) if and only if $|\alpha| < 1$. (In fact we could expand it for $|\alpha| > 1$ using $(1 - \alpha B)^{-1} = -(\alpha B)^{-1}(1 - \alpha^{-1}B^{-1})^{-1}$. However, this would give an expansion in terms of future values since B^{-1} is effectively a forward shift. This would not therefore be of much use. We will not point out this qualification in future.)

If $(1 - \alpha B)$ is invertible, then we can write:

$$\begin{aligned} X_t - \mu &= (1 - \alpha B)^{-1} e_t = (1 + \alpha B + \alpha^2 B^2 + \dots) e_t \\ &= e_t + \alpha e_{t-1} + \alpha^2 e_{t-2} + \dots \end{aligned}$$

In other words:

$$X_t = \mu + \sum_{j=0}^{\infty} \alpha^j e_{t-j}$$

This representation makes it clear that X_t has expectation μ and variance equal to:

$$\sum_{j=0}^{\infty} \alpha^{2j} \sigma^2 = \frac{\sigma^2}{1 - \alpha^2}$$

if $|\alpha| < 1$.

This last step uses the formula for the sum of an infinite geometric progression.

So, in this case, the process does satisfy the conditions required for stationarity given above.

We have only looked at the mean and variance so far, however. We also need to look at the autocovariance function.

In order to deduce that X is stationary we also need to calculate the autocovariance function:

$$\begin{aligned}\gamma_k &= \text{cov}(X_t, X_{t+k}) = \sum_{j=0}^{\infty} \sum_{i=0}^{\infty} \alpha^i \alpha^j \text{cov}(e_{t-j}, e_{t+k-i}) \\ &= \sum_{j=0}^{\infty} \sigma^2 \alpha^{2j+k} = \alpha^k \sum_{j=0}^{\infty} \sigma^2 \alpha^{2j} = \alpha^k \gamma_0\end{aligned}$$

The double sum has been simplified here by noting that the covariance will be zero unless the subscripts $t-j$ and $t+k-i$ are equal, ie unless $i=j+k$. In this case the covariance equals σ^2 . So in the sum over i , we only include the term for $i=j+k$.

We have also used the formula $\gamma_0 = \text{var}(X_t) = \sum_{j=0}^{\infty} \alpha^{2j} \sigma^2 = \frac{\sigma^2}{1-\alpha^2}$ from before.

This is independent of t , and thus a stationary process exists as long as $|\alpha| < 1$.

It is worth introducing here a method of more general utility for calculating autocovariance functions. From (13.1) we have, assuming that X is stationary:

$$\begin{aligned}\gamma_k &= \text{cov}(X_t, X_{t-k}) \\ &= \text{cov}(\mu + \alpha(X_{t-1} - \mu) + e_t, X_{t-k}) \\ &= \alpha \text{cov}(X_{t-1}, X_{t-k}) \\ &= \alpha \gamma_{k-1}\end{aligned}$$

implying that:

$$\gamma_k = \alpha^k \gamma_0 = \alpha^k \frac{\sigma^2}{1-\alpha^2}$$

and:

$$\rho_k = \frac{\gamma_k}{\gamma_0} = \alpha^k$$

for $k \geq 0$.

The partial autocorrelation function ϕ_k is given by:

$$\begin{aligned}\phi_1 &= \rho_1 = \alpha \\ \phi_2 &= \frac{\alpha^2 - \alpha^2}{1 - \alpha^2} = 0\end{aligned}$$

Indeed, since the best linear estimator of X_t given $X_{t-1}, X_{t-2}, X_{t-3}, \dots$ is just αX_{t-1} , the definition of the PACF implies that $\phi_k = 0$ for all $k > 1$. Notice the contrast with the ACF, which decreases geometrically towards 0.



The following lines in R generate the ACF and PACF functions for an $AR(1)$ model:

```
par(mfrow=c(1,2))

barplot(ARMAacf(ar=0.7,lag.max = 12)[-1],main = "ACF of AR(1)",
col="red")

barplot(ARMAacf(ar=0.7,lag.max = 12,pacf = TRUE),main = "PACF of
AR(1)",col="red")
```

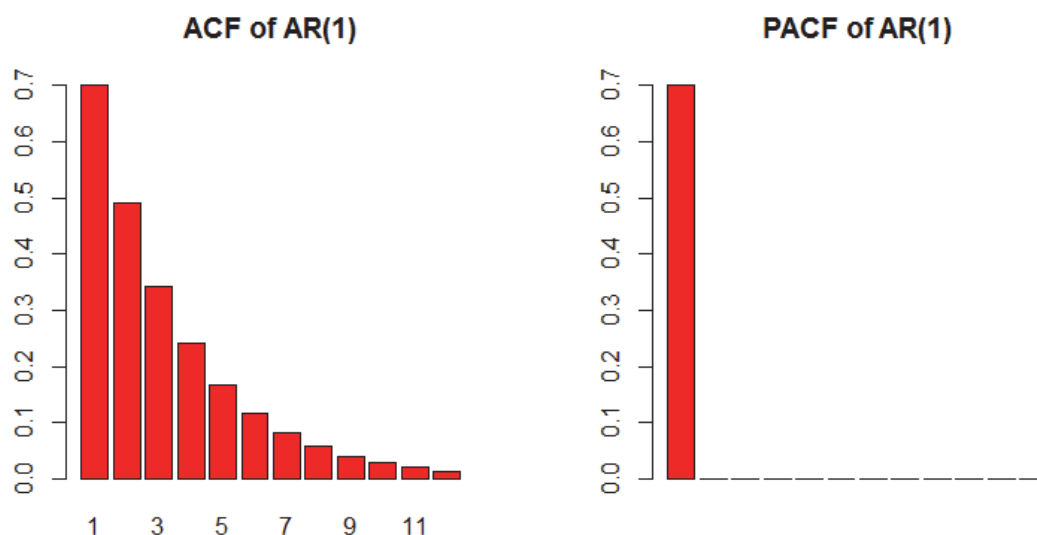


Figure 13.2: ACF and PACF of $AR(1)$ with $\alpha = 0.7$

Example

One of the well known applications of a univariate autoregressive model is the description of the evolution of the consumer price index $\{Q_t : t = 1, 2, 3, \dots\}$. The *force of inflation*, $r_t = \ln(Q_t/Q_{t-1})$, is assumed to follow the $AR(1)$ process:

$$r_t = \mu + \alpha(r_{t-1} - \mu) + e_t$$

One initial condition, the value for r_0 , is required for the complete specification of the model for the force of inflation r_t .

The process r_t is said to be *mean-reverting*, ie it has a long-run mean, and if it drifts away, then it tends to be dragged back towards it. In this case, the long-run mean is μ . The equation for r_t can be written in the form:

$$r_t - \mu = \alpha(r_{t-1} - \mu) + e_t$$

If we ignore the error term, which has a mean of zero, then this equation says that the difference between r and the long-run mean at time t , is α times the previous difference. In order to be mean-reverting, this distance must reduce, so we need $|\alpha| < 1$, as for stationarity. In fact, we probably wouldn't expect the force of inflation to be dragged to the other side of the mean, so a realistic model is likely to have $0 < \alpha < 1$.

3.4 The autoregressive model, $AR(p)$

The equation of the more general $AR(p)$ process is:

$$X_t = \mu + \alpha_1(X_{t-1} - \mu) + \alpha_2(X_{t-2} - \mu) + \dots + \alpha_p(X_{t-p} - \mu) + e_t \quad (13.4)$$

or, in terms of the backwards shift operator:

$$(1 - \alpha_1 B - \alpha_2 B^2 - \dots - \alpha_p B^p)(X_t - \mu) = e_t \quad (13.5)$$

As seen for $AR(1)$, there are some restrictions on the values of the α_j which are permitted if the process is to be stationary. In particular, we have the following result.



Condition for stationarity of an $AR(p)$ process (Result 13.2)

If the time series process X given by (13.4) is stationary then the roots of the equation:

$$1 - \alpha_1 z - \alpha_2 z^2 - \dots - \alpha_p z^p = 0$$

are all greater than 1 in absolute value.

(The polynomial $1 - \alpha_1 z - \alpha_2 z^2 - \dots - \alpha_p z^p$ is called the *characteristic polynomial* of the autoregression.)

The equation $1 - \alpha_1 z - \alpha_2 z^2 - \dots - \alpha_p z^p = 0$ is known as the *characteristic equation* of the process.

For an $AR(1)$ process, $X_t = \mu + \alpha(X_{t-1} - \mu) + e_t$, the characteristic equation is:

$$1 - \alpha z = 0$$

The root of this equation is $z = 1/\alpha$. So for an $AR(1)$ process to be stationary, we must have:

$$\frac{1}{|\alpha|} > 1$$

ie:

$$|\alpha| < 1$$

This is the same stationarity condition that we derived in Section 3.3.

It is important to realise what this result does *not* say. Although for an $AR(1)$ process we can look at the coefficient α and deduce stationarity if and only if $|\alpha| < 1$, we cannot do this for higher order autoregressive processes. For example, an $AR(2)$ process:

$$X_t = \mu + \alpha_1(X_{t-1} - \mu) + \alpha_2(X_{t-2} - \mu) + e_t$$

would not necessarily be stationary, just because $|\alpha_i| < 1$ for both $i=1$ and 2 . We would need to look at the roots of the characteristic polynomial.

We can prove the stationarity condition as follows.

Proof of Result 13.2

If X is stationary then its autocovariance function satisfies:

$$\gamma_k = \text{cov}(X_t, X_{t-k}) = \text{cov}\left(\sum_{j=1}^p \alpha_j X_{t-j} + e_t, X_{t-k}\right) = \sum_{j=1}^p \alpha_j \gamma_{k-j}$$

for $k \geq p$.

The μ 's are constant and do not therefore contribute to the covariance.

This is a p -th order difference equation with constant coefficients; it has a solution of the form:

$$\gamma_k = \sum_{j=1}^p A_j z_j^{-k}$$

for all $k \geq 0$, where $z_1, \dots, z_p$ are the p roots of the characteristic polynomial and $A_1, \dots, A_p$ are constants. (We will show this in a moment.) As X is purely indeterministic, we must have $\gamma_k \rightarrow 0$, which requires that $|z_j| > 1$ for each j .



Question

Show by substitution that $\gamma_k = \sum_{j=1}^p A_j z_j^{-k}$ is a solution of the given difference equation.

Solution

Substituting $\gamma_k = \sum_{j=1}^p A_j z_j^{-k}$ into the right-hand side of the difference equation $\gamma_k = \sum_{j=1}^p \alpha_j \gamma_{k-j}$

we get:

$$\sum_{j=1}^p \alpha_j \gamma_{k-j} = \sum_{j=1}^p \alpha_j \left(\sum_{i=1}^p A_i z_i^{-k+j} \right) = \sum_{i=1}^p A_i z_i^{-k} \left(\sum_{j=1}^p \alpha_j z_i^j \right)$$

By definition of z_i as a root of the characteristic equation, we have:

$$\sum_{j=1}^p \alpha_j z_i^j = 1$$

So:

$$\sum_{j=1}^p \alpha_j \gamma_{k-j} = \sum_{i=1}^p A_i z_i^{-k} = \gamma_k$$

as required.

The converse of Result 13.2 is also true (but the proof is not given here): if the roots of the characteristic polynomial are all greater than 1 in absolute value, then it is possible to construct a stationary process X satisfying (13.4). In order for an arbitrary process X satisfying (13.4) to be stationary, the variances and covariances of the initial values $X_0, X_{-1}, \dots, X_{-p+1}$ must also be equal to the appropriate values.

Although we do not give a formal proof, we will provide another way of thinking about this result.

Recall that in the $AR(1)$ case we said that the process turned out to be stationary if and only if X_t could be written as a (convergent) sum of white noise terms. Equivalently, if we start from the equation:

$$(1 - \alpha B)(X_t - \mu) = e_t$$

then the process is stationary if and only if we can invert the term $(1 - \alpha B)$, since this is the case if and only if $|\alpha| < 1$.

Analogously, we can write an $AR(p)$ process in the form:

$$\left(1 - \frac{B}{z_1}\right) \left(1 - \frac{B}{z_2}\right) \cdots \left(1 - \frac{B}{z_p}\right) (X_t - \mu) = e_t$$

where $z_1, z_2, \dots, z_p$ are the p (possibly complex) roots of the characteristic polynomial. In other words, the characteristic polynomial factorises as:

$$1 - \alpha_1 z - \alpha_2 z^2 - \cdots - \alpha_p z^p = \left(1 - \frac{z}{z_1}\right) \left(1 - \frac{z}{z_2}\right) \cdots \left(1 - \frac{z}{z_p}\right)$$

It follows that in order to write X_t in terms of the process e , we need to be able to invert all p of the factors $\left(1 - \frac{B}{z_i}\right)$. This will be the case if and only if $|z_i| > 1$ for all $i = 1, 2, \dots, p$.



Question

Determine the characteristic polynomial of the process defined by the equation:

$$X_t = 5 - 2(X_{t-1} - 5) + 3(X_{t-2} - 5) + e_t$$

and calculate its roots. Hence comment on the stationarity of the process.

Solution

We first rearrange the equation for the process so that all the X terms appear on the same side. Doing this we obtain:

$$X_t + 2X_{t-1} - 3X_{t-2} = e_t$$

We now replace X_{t-s} by z^s , ie we replace X_t by 1, X_{t-1} by z , and X_{t-2} by z^2 . So the characteristic polynomial is $1 + 2z - 3z^2$.

This polynomial can be factorised as $(1 + 3z)(1 - z)$, so its roots are $-\frac{1}{3}$ and 1.

This shows that the process is not stationary.

There is no requirement to use the letter z (or indeed any particular letter) when writing down the characteristic polynomial. The letter λ is often used instead.



Question

Given that $\lambda = 2$ is a root of the characteristic equation of the process:

$$X_n = \frac{11}{6}X_{n-1} - X_{n-2} + \frac{1}{6}X_{n-3} + e_n$$

calculate the other roots and classify the process as $I(d)$.

Solution

The process can be written in the form:

$$X_n - \frac{11}{6}X_{n-1} + X_{n-2} - \frac{1}{6}X_{n-3} = e_n$$

So that the characteristic equation is:

$$1 - \frac{11}{6}\lambda + \lambda^2 - \frac{1}{6}\lambda^3 = 0$$

We are given that $\lambda = 2$ is a root. So $(\lambda - 2)$ is a factor and hence:

$$1 - \frac{11}{6}\lambda + \lambda^2 - \frac{1}{6}\lambda^3 = (\lambda - 2)(a\lambda^2 + b\lambda + c)$$

where a , b and c are constants. The values of a , b and c can be determined in several ways, eg by comparing the coefficients on both sides of this equation, by long division of polynomials, or by synthetic division. We find that:

$$1 - \frac{11}{6}\lambda + \lambda^2 - \frac{1}{6}\lambda^3 = (\lambda - 2)\left(-\frac{1}{6}\lambda^2 + \frac{2}{3}\lambda - \frac{1}{2}\right) = -\frac{1}{6}(\lambda - 2)(\lambda - 3)(\lambda - 1)$$

So the other roots of the characteristic equation are $\lambda = 3$ and $\lambda = 1$.

The process is not stationary, since the characteristic equation has a root that is not strictly greater than 1 in magnitude.

It is easy to see that differencing the process once will eliminate the root of 1. The two remaining roots (ie 2 and 3) are both strictly greater than 1 in magnitude, so the differenced process is stationary. Hence X is $I(1)$.

Often exact values for the γ_k are required, entailing finding the values of the constants A_k . From (13.4) we have:

$$\text{cov}(X_t, X_{t-k}) = \alpha_1 \text{cov}(X_{t-1}, X_{t-k}) + \dots + \alpha_p \text{cov}(X_{t-p}, X_{t-k}) + \text{cov}(e_t, X_{t-k})$$

which can be re-expressed as:

$$\gamma_k = \alpha_1 \gamma_{k-1} + \alpha_2 \gamma_{k-2} + \dots + \alpha_p \gamma_{k-p} + \sigma^2 1_{\{k=0\}}$$

for $0 \leq k \leq p$. (These are known as the *Yule-Walker equations*.) Here the notation $1_{\{k=0\}}$ denotes an indicator function, taking the value 1 if $k = 0$, the value 0 otherwise.

For $p = 3$ we have 4 equations:

$$\gamma_3 = \alpha_1 \gamma_2 + \alpha_2 \gamma_1 + \alpha_3 \gamma_0$$

$$\gamma_2 = \alpha_1 \gamma_1 + \alpha_2 \gamma_0 + \alpha_3 \gamma_1$$

$$\gamma_1 = \alpha_1 \gamma_0 + \alpha_2 \gamma_1 + \alpha_3 \gamma_2$$

$$\gamma_0 = \alpha_1 \gamma_1 + \alpha_2 \gamma_2 + \alpha_3 \gamma_3 + \sigma^2$$

The second and third of these equations are sufficient to deduce γ_2 and γ_1 in terms of γ_0 , which is all that is required to find ρ_2 and ρ_1 . The first and fourth of the equations are needed when the values of the γ_k are to be found explicitly.

The PACF, $\{\phi_k : k \geq 1\}$, of the $AR(p)$ process can be calculated from the defining equations, but is not memorable. In particular, the first three equations above can be written in terms of ρ_1, ρ_2, ρ_3 and the resulting solution of α_3 as a function of ρ_1, ρ_2, ρ_3 is the expression of ϕ_3 . The same idea applies to all values of k , so that ϕ_k is the solution of α_k in a system of k linear equations, including those for $\phi_1 = \rho_1$ and $\phi_2 = \frac{\rho_2 - \rho_1^2}{1 - \rho_1^2}$ that we have seen before.

It is important to note, though, that $\phi_k = 0$ for all $k > p$.

This result is worth repeating.



Behaviour of the PACF for an $AR(p)$ process

For an $AR(p)$ process:

$$\phi_k = 0 \text{ for } k > p$$

This property of the PACF is characteristic of autoregressive processes and forms the basis of the most frequently used test for determining whether an $AR(p)$ model fits the data. It would be difficult to base a test on the ACF as the ACF of an autoregressive process is a sum of geometrically decreasing components. (See [Section 2](#) in Chapter 14.)



Question

Give a derivation of the equation:

$$\gamma_0 = \alpha_1 \gamma_1 + \alpha_2 \gamma_2 + \alpha_3 \gamma_3 + \sigma^2$$

Solution

The autocovariance at lag 0 is:

$$\gamma_0 = \text{cov}(X_t, X_t)$$

Expanding the LHS only (which will always be our approach when determining the autocovariance function of an autoregressive process), and remembering that the covariance is unaffected by the mean μ , we see that:

$$\gamma_0 = \text{cov}(\alpha_1 X_{t-1} + \alpha_2 X_{t-2} + \alpha_3 X_{t-3} + e_t, X_t)$$

Now, using the properties of covariance:

$$\begin{aligned} \gamma_0 &= \alpha_1 \text{cov}(X_{t-1}, X_t) + \alpha_2 \text{cov}(X_{t-2}, X_t) + \alpha_3 \text{cov}(X_{t-3}, X_t) + \text{cov}(e_t, X_t) \\ &= \alpha_1 \gamma_1 + \alpha_2 \gamma_2 + \alpha_3 \gamma_3 + \text{cov}(e_t, X_t) \end{aligned}$$

But:

$$\begin{aligned}\text{cov}(e_t, X_t) &= \text{cov}(e_t, \alpha_1 X_{t-1} + \alpha_2 X_{t-2} + \alpha_3 X_{t-3} + e_t) \\ &= \alpha_1 \text{cov}(e_t, X_{t-1}) + \alpha_2 \text{cov}(e_t, X_{t-2}) + \alpha_3 \text{cov}(e_t, X_{t-3}) + \text{cov}(e_t, e_t) \\ &= 0 + 0 + 0 + \sigma^2 = \sigma^2\end{aligned}$$

This is because $X_{t-1}, X_{t-2}, \dots$ are functions of past white noise terms, and e_t is independent of earlier values. So:

$$\gamma_0 = \alpha_1 \gamma_1 + \alpha_2 \gamma_2 + \alpha_3 \gamma_3 + \sigma^2$$

We will now derive a formula for the ACF of an $AR(2)$ process. To do this we need to remember a little about difference equations. Some formulae relating to difference equations are given on page 4 of the *Tables*.



Question

A stationary $AR(2)$ process is defined by the equation:

$$X_t = \frac{5}{6} X_{t-1} - \frac{1}{6} X_{t-2} + e_t$$

Determine the values of ρ_k and ϕ_k for $k = 1, 2, 3, \dots$.

Solution

We do not actually need to calculate γ_0 in order to find the ACF. This is always the case for an autoregressive process.

By definition:

$$\rho_0 = \frac{\gamma_0}{\gamma_0} = 1$$

The autocovariance at lag 1 is:

$$\begin{aligned}\gamma_1 &= \text{cov}(X_t, X_{t-1}) \\ &= \text{cov}\left(\frac{5}{6} X_{t-1} - \frac{1}{6} X_{t-2} + e_t, X_{t-1}\right) \\ &= \frac{5}{6} \text{cov}(X_{t-1}, X_{t-1}) - \frac{1}{6} \text{cov}(X_{t-2}, X_{t-1}) + \text{cov}(e_t, X_{t-1}) \\ &= \frac{5}{6} \gamma_0 - \frac{1}{6} \gamma_1\end{aligned}$$

Rearranging gives:

$$\frac{7}{6} \gamma_1 = \frac{5}{6} \gamma_0$$

So:

$$\gamma_1 = \frac{5}{7}\gamma_0 \quad \text{and} \quad \rho_1 = \frac{\gamma_1}{\gamma_0} = \frac{5}{7}$$

Similarly, the autocovariance at lag 2 is:

$$\begin{aligned} \gamma_2 &= \text{cov}(X_t, X_{t-2}) \\ &= \text{cov}\left(\frac{5}{6}X_{t-1} - \frac{1}{6}X_{t-2} + e_t, X_{t-2}\right) \\ &= \frac{5}{6}\text{cov}(X_{t-1}, X_{t-2}) - \frac{1}{6}\text{cov}(X_{t-2}, X_{t-2}) + \text{cov}(e_t, X_{t-2}) \\ &= \frac{5}{6}\gamma_1 - \frac{1}{6}\gamma_0 \end{aligned}$$

Using the fact that $\gamma_1 = \frac{5}{7}\gamma_0$, we have:

$$\gamma_2 = \frac{5}{6}\left(\frac{5}{7}\gamma_0\right) - \frac{1}{6}\gamma_0 = \frac{3}{7}\gamma_0 \quad \text{and} \quad \rho_2 = \frac{\gamma_2}{\gamma_0} = \frac{3}{7}$$

In general, for $k \geq 2$, we have:

$$\rho_k = \frac{5}{6}\rho_{k-1} - \frac{1}{6}\rho_{k-2}$$

We can solve this second order difference equation. The characteristic equation (for the difference equation, which is unfortunately slightly different to the characteristic equation for the process) is:

$$\lambda^2 - \frac{5}{6}\lambda + \frac{1}{6} = \left(\lambda - \frac{1}{2}\right)\left(\lambda - \frac{1}{3}\right) = 0$$

Using the formula on page 4 of the *Tables*, the general solution of this difference equation is of the form:

$$\rho_k = A\left(\frac{1}{2}\right)^k + B\left(\frac{1}{3}\right)^k$$

In order to find the solution we want, we need to use two boundary conditions to determine the two constants. We know that $\rho_0 = 1$ and $\rho_1 = \frac{5}{7}$. So:

$$\rho_0 = A + B = 1$$

$$\rho_1 = \frac{1}{2}A + \frac{1}{3}B = \frac{5}{7}$$

Solving these equations gives $A = \frac{16}{7}$ and $B = -\frac{9}{7}$.

The autocorrelation function is therefore:

$$\rho_k = \frac{16}{7}\left(\frac{1}{2}\right)^k - \frac{9}{7}\left(\frac{1}{3}\right)^k$$

We are also asked for the partial autocorrelation function. Using the formulae on page 40 of the *Tables*:

$$\phi_1 = \rho_1 = \frac{5}{7}$$

$$\phi_2 = \frac{\rho_2 - \rho_1^2}{1 - \rho_1^2} = -\frac{1}{6}$$

Also, since this is an $AR(2)$ process:

$$\phi_k = 0 \text{ for } k = 3, 4, 5, \dots$$

3.5 The first-order moving average model, $MA(1)$

A first-order moving average process, denoted $MA(1)$, is a process given by:

$$X_t = \mu + e_t + \beta e_{t-1}$$

The mean of this process is $\mu_t = \mu$.

The variance and autocovariance are:

$$\gamma_0 = \text{var}(e_t + \beta e_{t-1}) = (1 + \beta^2)\sigma^2$$

$$\gamma_1 = \text{cov}(e_t + \beta e_{t-1}, e_{t-1} + \beta e_{t-2}) = \beta\sigma^2$$

$$\gamma_k = 0 \text{ for } k > 1$$

Hence the ACF of the $MA(1)$ process is:

$$\rho_0 = 1$$

$$\rho_1 = \frac{\beta}{1 + \beta^2}$$

$$\rho_k = 0 \text{ for } k > 1$$



Question

Show that the moving average process $X_n = Z_n + \beta Z_{n-1}$ is weakly stationary, where Z_n is a white noise process with mean μ and variance σ^2 .

Solution

The mean is constant since $E(X_n) = (1 + \beta)\mu$.

For the covariance:

$$\text{cov}(X_n, X_n) = \text{cov}(Z_n + \beta Z_{n-1}, Z_n + \beta Z_{n-1}) = (1 + \beta^2)\sigma^2$$

or alternatively:

$$\text{var}(X_n) = \text{var}(Z_n + \beta Z_{n-1}) = (1 + \beta^2)\sigma^2$$

and:

$$\text{cov}(X_n, X_{n-1}) = \text{cov}(Z_n + \beta Z_{n-1}, Z_{n-1} + \beta Z_{n-2}) = \text{cov}(\beta Z_{n-1}, Z_{n-1}) = \beta\sigma^2$$

$$\text{cov}(X_n, X_{n-2}) = \text{cov}(Z_n + \beta Z_{n-1}, Z_{n-2} + \beta Z_{n-3}) = 0$$

In fact, the covariance at higher lags remains 0 since there is no overlap between the Z 's. The covariances at the corresponding negative lags are the same.

Since none of these expressions depends on n , it follows that the process is weakly stationary.

An $MA(1)$ process is stationary regardless of the values of its parameters. The parameters are nevertheless usually constrained by imposing the condition of *invertibility*. This may be explained as follows.

It is possible to have two distinct $MA(1)$ models with identical ACFs: consider, for example,

$$\beta = 0.5 \text{ and } \beta = 2, \text{ both of which have } \rho_1 = \frac{\beta}{1 + \beta^2} = 0.4.$$

The defining equation of the $MA(1)$ may be written in terms of the backwards shift operator:

$$X - \mu = (1 + \beta B)e \tag{13.6}$$

In many circumstances an autoregressive model is more convenient than a moving average model.

We may rewrite (13.6) as:

$$(1 + \beta B)^{-1}(X - \mu) = e$$

and use the standard expansion of $(1 + \beta B)^{-1} = 1 - \beta B + \beta^2 B^2 - \beta^3 B^3 + \dots$ to give:

$$X_t - \mu - \beta(X_{t-1} - \mu) + \beta^2(X_{t-2} - \mu) - \beta^3(X_{t-3} - \mu) + \dots = e_t$$

The expansion referred to here is given on page 2 of the *Tables*.

The original moving average model has therefore been transformed into an autoregression of infinite order. But this procedure is only valid if the sum on the left-hand side is convergent, in other words if $|\beta| < 1$. When this condition is satisfied the $MA(1)$ is called *invertible*. Although more than one MA process may share a given ACF, at most one of the processes will be invertible.

We might want to know the historic values of the white noise process. Although the values $\{x_0, x_1, \dots, x_t\}$ can be observed, and are therefore known at time t , the values of the white noise process $\{e_0, e_1, \dots, e_t\}$ are not. Can we obtain the unknown e values from the known x values? The answer is yes, in theory, if and only if the process is invertible, since we can then write the value e_t in terms of the x 's, as above. In practice, we wouldn't actually have an infinite history of x values, but since the coefficients of the x 's get smaller as we go back in time, for an invertible process, the contribution of the values before time 1, say, will be negligible. We can make this more precise, as in the following question.



Question

Show that the process $X_n = \mu + e_n + \beta e_{n-1}$ may be inverted as follows:

$$e_n = (-\beta)^n e_0 + \sum_{i=0}^{n-1} (-\beta)^i (x_{n-i} - \mu)$$

Solution

A simple algebraic rearrangement shows that $X_n = \mu + e_n + \beta e_{n-1}$ can be rewritten as $e_n = X_n - \mu - \beta e_{n-1}$. Now using an iterative procedure:

$$\begin{aligned} e_n &= X_n - \mu - \beta e_{n-1} \\ &= X_n - \mu - \beta(X_{n-1} - \mu - \beta e_{n-2}) \\ &\vdots \\ &= (X_n - \mu) - \beta(X_{n-1} - \mu) + \beta^2(X_{n-2} - \mu) + \dots + (-\beta)^{n-1}(X_1 - \mu) + (-\beta)^n e_0 \end{aligned}$$

Notice that, as n gets large, the dependence of e_n on e_0 will be small provided $|\beta| < 1$.

The condition for a $MA(1)$ process to be invertible is similar to the condition that an $AR(1)$ process is stationary. An $AR(1)$ process is stationary if and only if the process X can be written explicitly in terms of the process e . The invertibility condition ensures that the white noise process e can be written in terms of the X process. This relationship generalises to $AR(p)$ and $MA(q)$ processes, as we will see shortly.

It is possible, at the cost of considerable effort, to calculate the PACF of the $MA(1)$, giving:

$$\phi_k = (-1)^{k+1} \frac{(1-\beta^2)\beta^k}{1-\beta^{2(k+1)}}$$

This formula can be found on page 41 of the *Tables*.

This decays approximately geometrically as $k \rightarrow \infty$, highlighting the way in which the ACF and PACF are complementary: the PACF of a $MA(1)$ behaves like the ACF of an $AR(1)$, and the PACF of an $AR(1)$ behaves like the ACF of a $MA(1)$.

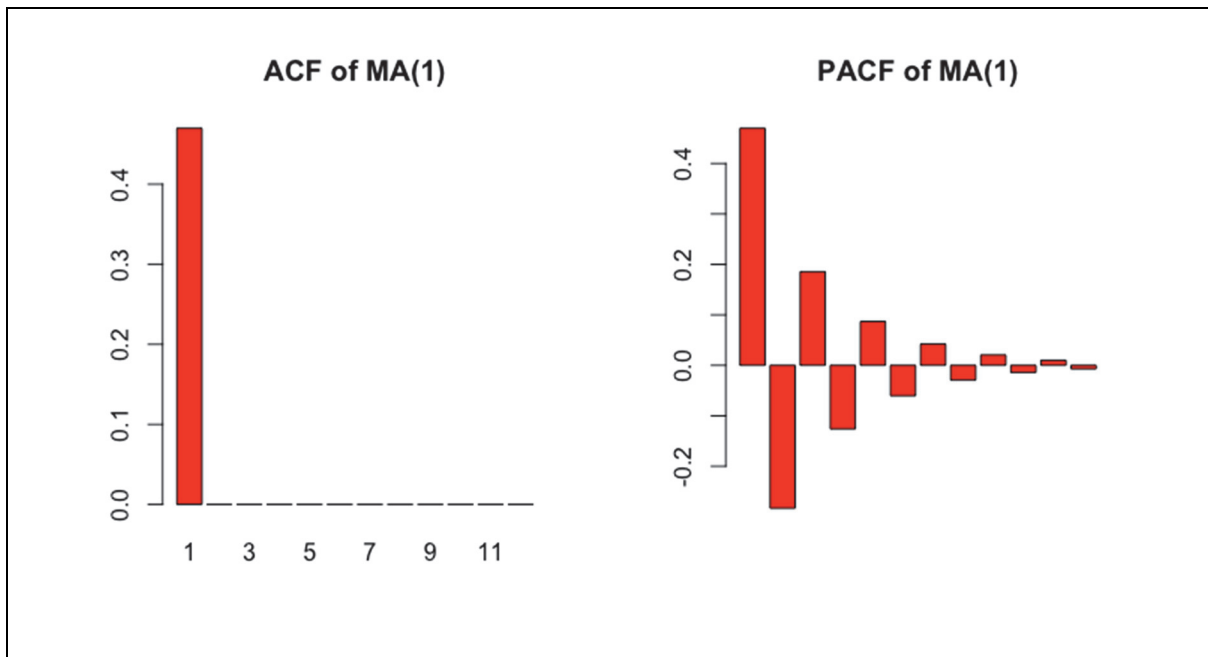


Figure 13.3: ACF and PACF of $MA(1)$ with $\beta = 0.7$

3.6 The moving average model, $MA(q)$

The defining equation of the general q th order moving average is, in backwards shift notation:

$$X - \mu = (1 + \beta_1 B + \beta_2 B^2 + \dots + \beta_q B^q) e$$

In other words, it is:

$$X_t - \mu = e_t + \beta_1 e_{t-1} + \beta_2 e_{t-2} + \dots + \beta_q e_{t-q}$$

Moving average processes are always stationary, as they are a linear combination of white noise, which is itself stationary.

Recall that for a stationary $AR(p)$ process, X_t can be expressed as an (infinite and convergent) sum of white noise terms. This means that any stationary autoregressive process can be considered to be a moving average of infinite order. However, by a moving average process we will usually mean one of finite order.

The autocovariance function is easier to find than in the case of $AR(p)$:

$$\gamma_k = \sum_{i=0}^q \sum_{j=0}^q \beta_i \beta_j E(e_{t-i} e_{t-j-k}) = \sigma^2 \sum_{i=0}^{q-k} \beta_i \beta_{i+k}$$

provided $k \leq q$. (Here β_0 denotes 1.)

Note that $\text{cov}(e_i, e_j) = E(e_i e_j) - E(e_i)E(e_j) = E(e_i e_j) = 0$ for $i \neq j$ (since the random variables e_t have 0 mean and are uncorrelated), and $\text{cov}(e_i, e_i) = \text{var}(e_i) = E(e_i^2) = \sigma^2$. So, in the double sum above, the only non-zero terms will be where the subscripts of e_{t-i} and e_{t-j-k} match, ie when $i = j + k$. This means that we can simplify the double sum by writing everything in terms of j . We need to get the limits right for j , which cannot go above $q - k$ because $i = j + k$ and i only goes up to q . So we get:

$$\gamma_k = \sum_{j=0}^{q-k} \beta_{j+k} \beta_j \sigma^2$$

This matches the formula above, except that i has been used in place of j .

For $k > q$ it is obvious that $\gamma_k = 0$. Just as autoregressive processes are characterised by the property that the partial ACF is equal to zero for sufficiently large k , moving average processes are characterised by the property that the ACF is equal to zero for sufficiently large k .

The importance of this observation will become apparent in Section 2 of [Chapter 14](#). We will look at an explicit case of this result to make things clearer.



Question

Calculate γ_k , $k = 0, 1, 2, 3, \dots$ for the process:

$$X_n = 3 + e_n - e_{n-1} + 0.25e_{n-2}$$

where e_n is a white noise process with mean 0 and variance 1.

Solution

We have:

$$\begin{aligned} \gamma_0 &= \text{cov}(X_n, X_n) = \text{var}(X_n) \\ &= \text{var}(e_n - e_{n-1} + 0.25e_{n-2}) \\ &= \text{var}(e_n) + (-1)^2 \text{var}(e_{n-1}) + 0.25^2 \text{var}(e_{n-2}) \\ &= 1 + 1 + 0.0625 \\ &= 2.0625 \end{aligned}$$

$$\begin{aligned}
\gamma_1 &= \text{cov}(X_n, X_{n-1}) \\
&= \text{cov}(e_n - e_{n-1} + 0.25e_{n-2}, e_{n-1} - e_{n-2} + 0.25e_{n-3}) \\
&= \text{cov}(-e_{n-1}, e_{n-1}) + \text{cov}(0.25e_{n-2}, -e_{n-2}) \\
&= -1 - 0.25 \\
&= -1.25 \\
\gamma_2 &= \text{cov}(X_n, X_{n-2}) \\
&= \text{cov}(e_n - e_{n-1} + 0.25e_{n-2}, e_{n-2} - e_{n-3} + 0.25e_{n-4}) \\
&= \text{cov}(0.25e_{n-2}, e_{n-2}) \\
&= 0.25 \\
\gamma_3 &= \text{cov}(X_n, X_{n-3}) \\
&= \text{cov}(e_n - e_{n-1} + 0.25e_{n-2}, e_{n-3} - e_{n-4} + 0.25e_{n-5}) \\
&= 0
\end{aligned}$$

The covariance at higher lags is also 0 since there is no overlap between the e 's.

In the solution above, we have expanded the terms on both sides of the covariance expression. This will always be our strategy when calculating the autocovariance function for a moving average series. For all other types of series, we just expand the term on the LHS of the covariance expression.

We said above that an $MA(1)$ process $X_t = \mu + e_t + \beta e_{t-1}$ is invertible if $|\beta| < 1$, and we drew the analogy with an $AR(1)$ process being stationary. The same goes for this more general case. Recall that an $AR(p)$ process is stationary if and only if the roots of the characteristic equation are all strictly greater than 1 in magnitude.

For an $MA(q)$ process we have:

$$X_t - \mu = (1 + \beta_1 B + \beta_2 B^2 + \dots + \beta_q B^q) e_t$$

The equation:

$$1 + \beta_1 z + \beta_2 z^2 + \dots + \beta_q z^q = 0$$

can be used to determine invertibility. This can be thought of as the characteristic equation of the white noise terms.



Condition for invertibility of a $MA(q)$ process

The process X defined by the equation:

$$X_t - \mu = e_t + \beta_1 e_{t-1} + \beta_2 e_{t-2} + \cdots + \beta_q e_{t-q}$$

is invertible if and only if the roots of the equation:

$$1 + \beta_1 z + \beta_2 z^2 + \cdots + \beta_q z^q = 0$$

are all strictly greater than 1 in magnitude.

This is equivalent to saying that the value e_t can be written explicitly as a (convergent) sum of X values.

Again, λ is often used instead of z in the characteristic equation.

It follows that in the same way as a stationary autoregression can be thought of as a moving average of infinite order, so a moving average can be thought of as an autoregression of infinite order.



Question

Determine whether the process $X_t = 2 + e_t - 5e_{t-1} + 6e_{t-2}$ is invertible.

Solution

The equation $1 - 5\lambda + 6\lambda^2 = (1 - 2\lambda)(1 - 3\lambda) = 0$ has roots $1/3$ and $1/2$, so the process is not invertible.

Although there may be many moving average processes with the same ACF, at most one of them is invertible, since no two invertible processes have the same autocorrelation function. Moving average models fitted to data by statistical packages will always be invertible.

3.7 The autoregressive moving average process, $ARMA(p, q)$

A combination of the moving average and autoregressive models, an ARMA model includes direct dependence of X_t on both past values of X and present and past values of e .

The defining equation is:

$$X_t = \mu + \alpha_1(X_{t-1} - \mu) + \cdots + \alpha_p(X_{t-p} - \mu) + e_t + \beta_1 e_{t-1} + \cdots + \beta_q e_{t-q}$$

or, in backwards shift operator notation:

$$(1 - \alpha_1 B - \cdots - \alpha_p B^p)(X - \mu) = (1 + \beta_1 B + \cdots + \beta_q B^q)e$$

This might also be written:

$$\phi(B)(X_n - \mu) = \theta(B)e_n$$

where $\theta(B)$ and $\phi(B)$ are polynomials of degrees q and p , respectively.

If $\theta(B)$ and $\phi(B)$ have any factors (*ie* roots) in common, then the defining relation could be simplified. For example, we may have a stationary $ARMA(1,1)$ process defined by $(1 - \alpha B)X_n = (1 - \alpha B)e_n$ with $|\alpha| < 1$. Since the factors $(1 - \alpha B)$ are invertible, we can multiply both sides of the equation by $(1 - \alpha B)^{-1}$ to see that $X_n = e_n$. So this process would actually be classified as $ARMA(0,0)$.

In general, it would be assumed that the polynomials $\theta(B)$ and $\phi(B)$ have no common roots.

Autoregressive and moving average processes are special cases of ARMA processes. $AR(p)$ is the same as $ARMA(p,0)$. $MA(q)$ is the same as $ARMA(0,q)$.

To check the stationarity of an ARMA process, we just need to examine the autoregressive part. The moving average part (which involves the white noise terms) is always stationary. The test is the same as for an autoregressive process – we need to determine the roots of the characteristic equation formed by the X terms. The process is stationary if and only if all the roots are strictly greater than 1 in magnitude.

Similarly, we can check for invertibility by examining the roots of the characteristic equation that is obtained from the white noise terms. The process is invertible if and only if all the roots are strictly greater than 1 in magnitude.

Neither the ACF nor the PACF of the ARMA process eventually becomes equal to zero. This makes it more difficult to identify an ARMA model than either a pure autoregression or a pure moving average.

Theoretically, both the ACF and PACF of a stationary ARMA process will tend towards 0 for large lags, but neither will have a cut off property.

It is possible to calculate the ACF by a method similar to the method employing the Yule-Walker equations for the ACF of an autoregression.

We will show that the autocorrelation function of the stationary zero-mean $ARMA(1,1)$ process:

$$X_t = \alpha X_{t-1} + e_t + \beta e_{t-1} \quad (13.7)$$

is given by:

$$\rho_1 = \frac{(1 + \alpha\beta)(\alpha + \beta)}{(1 + \beta^2 + 2\alpha\beta)}$$

$$\rho_k = \alpha^{k-1} \rho_1, \quad k = 2, 3, \dots$$

These results can be obtained from the formula for ρ_k given on page 40 of the *Tables*.

Figure 13.1 in Section 2.5 shows the ACF and PACF values of such a process with $\alpha = 0.7$ and $\beta = 0.5$.

Before we tackle the Yule-Walker equations, we need a couple of preliminary results.

Using equation (13.7), ie:

$$X_t = \alpha X_{t-1} + e_t + \beta e_{t-1}$$

we have:

$$\begin{aligned}\text{cov}(X_t, e_t) &= \alpha \text{cov}(X_{t-1}, e_t) + \text{cov}(e_t, e_t) + \beta \text{cov}(e_{t-1}, e_t) \\ &= \sigma^2\end{aligned}$$

since e_t is independent of both e_{t-1} and X_{t-1} .

Similarly:

$$\begin{aligned}\text{cov}(X_t, e_{t-1}) &= \alpha \text{cov}(X_{t-1}, e_{t-1}) + \text{cov}(e_t, e_{t-1}) + \beta \text{cov}(e_{t-1}, e_{t-1}) \\ &= (\alpha + \beta)\sigma^2\end{aligned}$$

This enables us to deduce the autocovariance function of X . Again from (13.7):

$$\text{cov}(X_t, X_t) = \alpha \text{cov}(X_{t-1}, X_t) + \text{cov}(e_t, X_t) + \beta \text{cov}(e_{t-1}, X_t)$$

$$\text{cov}(X_t, X_{t-1}) = \alpha \text{cov}(X_{t-1}, X_{t-1}) + \text{cov}(e_t, X_{t-1}) + \beta \text{cov}(e_{t-1}, X_{t-1})$$

and, for $k > 1$:

$$\text{cov}(X_t, X_{t-k}) = \alpha \text{cov}(X_{t-1}, X_{t-k}) + \text{cov}(e_t, X_{t-k}) + \beta \text{cov}(e_{t-1}, X_{t-k})$$

So:

$$\gamma_0 = \alpha\gamma_1 + (1 + \alpha\beta + \beta^2)\sigma^2$$

$$\gamma_1 = \alpha\gamma_0 + \beta\sigma^2$$

$$\gamma_k = \alpha\gamma_{k-1}$$

The solution is:

$$\gamma_0 = \frac{1 + 2\alpha\beta + \beta^2}{1 - \alpha^2} \sigma^2$$

$$\gamma_1 = \frac{(\alpha + \beta)(1 + \alpha\beta)}{1 - \alpha^2} \sigma^2$$

$$\gamma_k = \alpha^{k-1} \gamma_1 \quad k = 2, 3, \dots$$

assuming that the process is stationary, ie that $|\alpha| < 1$.

Hence:

$$\rho_1 = \frac{\gamma_1}{\gamma_0} = \frac{(1 + \alpha\beta)(\alpha + \beta)}{(1 + 2\alpha\beta + \beta^2)}$$

and:

$$\rho_k = \frac{\gamma_k}{\gamma_0} = \frac{\alpha^{k-1}\gamma_1}{\gamma_0} = \alpha^{k-1}\rho_1, \quad k = 2, 3, \dots$$



Question

Show that the process $12X_t = 10X_{t-1} - 2X_{t-2} + 12e_t - 11e_{t-1} + 2e_{t-2}$ is both stationary and invertible.

Solution

We start by rewriting the process so that all the X 's are on the same side and all the e 's are on the same side:

$$12X_t - 10X_{t-1} + 2X_{t-2} = 12e_t - 11e_{t-1} + 2e_{t-2}$$

The characteristic equation of the AR part is $12 - 10\lambda + 2\lambda^2 = (2\lambda - 4)(\lambda - 3) = 0$, which has roots 2 and 3. The process is therefore stationary.

The characteristic equation of the MA part is $12 - 11\lambda + 2\lambda^2 = (2\lambda - 3)(\lambda - 4) = 0$. The roots of this equation are 1.5 and 4. The process is therefore invertible.

3.8 Modelling non-stationary processes: the ARIMA model

This is the most general class of models we will consider. They lie at the heart of the Box-Jenkins approach to modelling time series. In order to understand the rationale underlying the definition, it will be useful to give a brief introduction to the Box-Jenkins method; a more detailed discussion will be given in [Chapter 14](#).

Suppose we are given some time series data x_n , where n varies over some finite range. If we want to model the data, then we would expect to take sample statistics, in particular the sample autocorrelation function, sample partial autocorrelation function and sample mean; these will be discussed in more detail in [Chapter 14](#). The modelling process would then involve finding a stochastic process with similar characteristics. In the Box-Jenkins approach, the model is picked from the $ARMA(p, q)$ class. However, the theoretical counterparts of the autocorrelation and partial autocorrelation functions are only defined for stationary series. The upshot of this is that we can only *directly* apply these methods if the original data values are stationary.

However, we can get around this problem as follows. First transform the data into a stationary form, which we will discuss in a moment. We can then model this stationary series as suggested above. Finally, we carry out the inverse transform on our model to obtain a model for the original series. The question remains as to what we mean by ‘transform’ and ‘inverse transform’.

The backward difference operator can turn a non-stationary series into a stationary one. For example, a random walk:

$$X_t = X_{t-1} + e_t$$

is non-stationary (as its characteristic equation is $1 - \lambda = 0$, which has a root of 1). However, the difference:

$$\nabla X_t = X_t - X_{t-1} = e_t$$

is just white noise, which is stationary.

For the moment we will assume that it's possible to turn the data set into a stationary series by repeated differencing. We may have to difference the series several times, the specific number usually being denoted by d .

Now assuming we've transformed our data into stationary form by differencing, we can model this series using a *stationary* $ARMA(p, q)$ process. The final step is to reverse the differencing procedure to obtain a model of the original series. The inverse process of differencing is integration since we must sum the differences to obtain the original series.



Question

From a data set $x_0, x_1, x_2, \dots, x_N$ the first order differences $w_i = x_i - x_{i-1}$ are calculated.

State the range of values of i and give an expression for x_j in terms of x_0 and the w 's.

Solution

The values of i are $1, 2, \dots, N$ and:

$$x_j = x_0 + \sum_{i=1}^j w_i$$

In many applications the process being modelled cannot be assumed stationary, but can reasonably be fitted by a model with stationary increments, that is, if the first difference of X , $Y = \nabla X$, is itself a stationary process.

A process X is called an $ARIMA(p, 1, q)$ process if X is non-stationary but the first difference of X is a stationary $ARMA(p, q)$ process.

We will now consider some examples.

Example 1

The simplest example of an ARIMA process is the random walk, $X_t = X_{t-1} + e_t$, which can be written $X_t = X_0 + \sum_{j=1}^t e_j$. The expectation of X_t is equal to $E[X_0]$ but the variance is $\text{var}(X_0) + t\sigma^2$, so that X is not itself stationary.

Here we are assuming (as we usually do) that the white noise process has zero mean.

The first difference, however, is given by:

$$Y_t = \nabla X_t = e_t$$

which certainly is stationary. Thus the random walk is an $ARIMA(0,1,0)$ process.

Example 2

Let Z_t denote the closing price of a share on day t . The evolution of Z is frequently described by the model:

$$Z_t = Z_{t-1} \exp(\mu + e_t)$$

By taking logarithms we see that this model is equivalent to an $I(1)$ model, since $Y_t = \ln Z_t$ satisfies the equation:

$$Y_t = \mu + Y_{t-1} + e_t$$

which is the defining equation of a random walk with drift because $Y_t = Y_0 + \mu t + \sum_{j=1}^t e_j$. The

model is based on the assumption that the daily returns $\ln(Z_t/Z_{t-1})$ are independent of the past prices $Z_0, Z_1, \dots, Z_{t-1}$.

Example 3

The logarithm of the consumer price index can be described by the $ARIMA(1,1,0)$ model:

$$(1-B)\ln Q_t = \mu + \alpha[(1-B)\ln Q_{t-1} - \mu] + e_t$$

When analysing the behaviour of an $ARIMA(p,1,q)$ model, the standard technique is to look at the first difference of the process and to perform the kind of analysis which is suitable for an ARMA model. Once complete, this can be used to provide predictions for the original, undifferenced, process.

$ARIMA(p,d,q)$ processes

In certain cases it may be considered desirable to continue beyond the first difference, if the process X is still not stationary after being differenced once. The notation extends in a natural way.



Definition of an *ARIMA* process

If X needs to be differenced at least d times in order to reduce it to stationarity and if the d th difference $Y = \nabla^d X$ is an *ARMA*(p, q) process, then X is termed an *ARIMA*(p, d, q) process.

In terms of the backwards shift operator, the equation of the *ARIMA*(p, d, q) process is:

$$(1 - \alpha_1 B - \dots - \alpha_p B^p)(1 - B)^d (X - \mu) = (1 + \beta_1 B + \dots + \beta_q B^q) e$$

An *ARIMA*(p, d, q) process is $I(d)$. We can think of the classification *ARIMA*(p, d, q) as:

$$AR(p)I(d)MA(q)$$

We now consider an example where we classify a time series as an *ARIMA* process.

To identify the values of p , d and q for which X is an *ARIMA*(p, d, q) process, where:

$$X_t = 0.6X_{t-1} + 0.3X_{t-2} + 0.1X_{t-3} + e_t - 0.25e_{t-1}$$

we can write the equation in terms of the backwards shift operator:

$$(1 - 0.6B - 0.3B^2 - 0.1B^3)X = (1 - 0.25B)e$$

We now check whether the polynomial on the left-hand side is divisible by $1 - B$; if so, factorise it out. We continue to do this until the remaining polynomial is not divisible by $1 - B$.

$$(1 - B)(1 + 0.4B + 0.1B^2)X = (1 - 0.25B)e$$

The model can now be seen to be *ARIMA*(2, 1, 1).

We should also check that the roots of the characteristic polynomial of ∇X_t , ie $1 + 0.4\lambda + 0.1\lambda^2$, are both strictly greater than 1 in magnitude. In fact, the roots are $-2 \pm i\sqrt{6}$. The magnitude of the complex number $a + bi$ is $\sqrt{a^2 + b^2}$. So the magnitude of $-2 + i\sqrt{6}$ is:

$$\sqrt{(-2)^2 + (\sqrt{6})^2} = \sqrt{4 + 6} = \sqrt{10}$$

The magnitude of $-2 - i\sqrt{6}$ is also $\sqrt{10}$. So both roots are strictly greater than 1 in magnitude.

Differencing once removes the factor of $(1 - B)$. Hence, the process ∇X_t is stationary, as required.

Alternatively, we could write the equation for the process as:

$$X_t - 0.6X_{t-1} - 0.3X_{t-2} - 0.1X_{t-3} = e_t - 0.25e_{t-1}$$

The characteristic equation of the AR part is:

$$1 - 0.6\lambda - 0.3\lambda^2 - 0.1\lambda^3 = 0$$

There is no simple formula for solving cubic equations, so we should start by checking whether 1 is a root of this equation. Setting $\lambda = 1$, we see that the left-hand side is:

$$1 - 0.6 - 0.3 - 0.1 = 0$$

So 1 is a root, and hence $(\lambda - 1)$ is a factor. The characteristic polynomial can therefore be written in the form:

$$(\lambda - 1)(a\lambda^2 + b\lambda + c)$$

We can determine the values of a , b and c by comparing coefficients, by long division of polynomials, or by synthetic division. We find that $a = 0.1$, $b = 0.4$ and $c = 1$, and the roots of the equation $0.1\lambda^2 + 0.4\lambda + 1 = 0$ are $-2 \pm i\sqrt{6}$ as stated above, and these are strictly greater than 1 in magnitude. Since the characteristic equation has one root of 1 and the other roots are strictly greater than 1 in magnitude, differencing once will give us a stationary process. So $d = 1$. There are two other roots, so $p = 2$. In addition, $q = 1$ since the moving average part is of order 1. Hence the process is $ARIMA(2,1,1)$.

Another alternative is to write the defining equation in terms of the differences.

The equation:

$$X_t = 0.6X_{t-1} + 0.3X_{t-2} + 0.1X_{t-3} + e_t - 0.25e_{t-1}$$

can be rearranged as:

$$X_t - 0.6X_{t-1} - 0.3X_{t-2} - 0.1X_{t-3} = e_t - 0.25e_{t-1}$$

or:

$$(X_t - X_{t-1}) + 0.4(X_{t-1} - X_{t-2}) + 0.1(X_{t-2} - X_{t-3}) = e_t - 0.25e_{t-1}$$

or:

$$\nabla X_t + 0.4\nabla X_{t-1} + 0.1\nabla X_{t-2} = e_t - 0.25e_{t-1}$$

The characteristic equation formed by the ∇X terms is:

$$0.1\lambda^2 + 0.4\lambda + 1 = 0$$

As we have already seen, the roots of this equation are $-2 \pm i\sqrt{6}$, and these both have a magnitude of $\sqrt{10}$. So ∇X is stationary $ARMA(2,1)$ and hence X is $ARIMA(2,1,1)$.

ARIMA models play a central role in the Box-Jenkins methodology, which aims to provide a consistent and unified framework for analysis and prediction using time series models. (See Section 3.1 of Chapter 14.)

3.9 The Markov property

As we saw in [Chapter 1](#), if the future development of a process can be predicted from its present state alone, without any reference to its past history, it possesses the Markov property. Stated precisely this reads:

$$P[X_t \in A \mid X_{s_1} = x_1, X_{s_2} = x_2, \dots, X_{s_n} = x_n, X_s = x] = P[X_t \in A \mid X_s = x]$$

for all times $s_1 < s_2 < \dots < s_n < s < t$, all states $x_1, x_2, \dots, x_n$ and x in S and all subsets A of S .

Recall from Section 1 that a time series process is defined as having a continuous state space. The necessity to work with subsets $A \subseteq S$ (rather than just having $X_t = a \in S$) is to cover these continuous state space cases. For these the probability that X_t takes on a particular value is zero. We therefore need to work with probabilities of X_t lying in some interval of S , or more generally in some subset.

A first-order autoregressive process possesses the Markov property, since the conditional distribution of X_{n+1} given all previous X_t depends only on X_n . This property does not apply, however, to higher-order autoregressive models.

Suppose X is an $AR(2)$. X does not possess the Markov property, since the conditional distribution of X_{n+1} given the history of X up until time n depends on X_{n-1} as well as on X_n . But let us define a vector-valued process $\underline{Y}$ by $\underline{Y}_t = (X_t, X_{t-1})^T$.

Vector-valued or multivariate processes will be studied in more detail in [Chapter 14](#). The superscript ' T ' here means transpose, ie we are thinking of $\underline{Y}$ as a column vector.

Given the whole history of the process X up until time n , the distribution of $\underline{Y}_{n+1}$ depends only on the values of X_n and X_{n-1} – in other words, on the value of $\underline{Y}_n$. This means that $\underline{Y}$ possesses the Markov property.

We can illustrate this as follows. The $AR(2)$ process:

$$X_n = \alpha_1 X_{n-1} + \alpha_2 X_{n-2} + e_n$$

can be written in matrix form as follows:

$$\begin{pmatrix} X_n \\ X_{n-1} \end{pmatrix} = \begin{pmatrix} \alpha_1 & \alpha_2 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} X_{n-1} \\ X_{n-2} \end{pmatrix} + \begin{pmatrix} e_n \\ 0 \end{pmatrix}$$

ie:

$$\underline{Y}_n = \begin{pmatrix} \alpha_1 & \alpha_2 \\ 1 & 0 \end{pmatrix} \underline{Y}_{n-1} + \begin{pmatrix} e_n \\ 0 \end{pmatrix}$$

We can see from the equation immediately above that future values of the process $\underline{Y}$ depend only on the current value, and not on the past history of the process.

Similarly, the $AR(3)$ process:

$$X_n = \alpha_1 X_{n-1} + \alpha_2 X_{n-2} + \alpha_3 X_{n-3} + e_n$$

can be written in matrix form as follows:

$$\begin{pmatrix} X_n \\ X_{n-1} \\ X_{n-2} \end{pmatrix} = \begin{pmatrix} \alpha_1 & \alpha_2 & \alpha_3 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} X_{n-1} \\ X_{n-2} \\ X_{n-3} \end{pmatrix} + \begin{pmatrix} e_n \\ 0 \\ 0 \end{pmatrix}$$

In general an $AR(p)$ does not possess the Markov property (for $p > 1$) but we may define a vector-valued process $\underline{Y}_t = (X_t, X_{t-1}, \dots, X_{t-p+1})^T$ which does.

You will recall from [Chapter 1](#) that a random walk possesses the Markov property. The discussion of the Markov property for autoregressions can be extended to include some *ARIMA* processes such as the random walk, which has already been shown to be an $ARIMA(0,1,0)$ process: an $ARIMA(p,d,0)$ process does not possess the Markov property (for $p+d > 1$) but we may define a vector-valued process $\underline{Y}_t = (X_t, X_{t-1}, \dots, X_{t-p-d+1})^T$ which does.

A moving average, or more generally an $ARIMA(p,d,q)$ process with $q > 0$, can never be Markov, since knowledge of the value of X_n , or of any finite collection

$(X_n, X_{n-1}, \dots, X_{n-q+1})^T$ will never be enough to deduce the value of e_n , on which the distribution of X_{n+1} depends. Since a moving average has been shown to be equivalent to an autoregression of infinite order, and since a p th order autoregression needs to be expressed as a p -dimensional vector in order to possess the Markov property, a moving average has no similar finite-dimensional Markov representation.



Question

Let $X_n = e_n + e_{n-1}$ be an $MA(1)$ process where $e_n \sim N(0,1)$. Determine whether this process has the Markov property.

Solution

The process is not Markov. We can see this intuitively as follows (although we could explicitly calculate some conditional probabilities to give a more formal proof).

The problem is really whether knowing the value of X_{n-2} in addition to X_{n-1} will help in predicting the value of X_n . Suppose, for example, that $X_{n-1} = 0$, ie $e_{n-1} + e_{n-2} = 0$. We cannot deduce the values of the e 's themselves.

However, the value X_{n-2} will give us extra information as to the likely value of e_{n-2} , which in turn gives us extra information about the value of e_{n-1} , which in turn gives us information about the value of X_n .

For example, X_{n-2} might be very large and positive, which would indicate that e_{n-2} is more likely to be positive, which would indicate that e_{n-1} is more likely to be negative, which would indicate that X_n is more likely to be negative.

Chapter 13 Summary

Univariate time series

A univariate time series is a sequence of observations $\{X_t\}$ recorded at regular intervals. The state space is continuous but the time set is discrete.

Such series may follow a pattern to some extent, for example possessing a trend or seasonal component, as well as having random factors. The aim is to construct a model to fit a set of past data in order to forecast future values of the series.

Stationarity

It is easier (more efficient) to construct a model if the time series is stationary.

A time series is said to be stationary, or strictly stationary, if the joint distributions of $\{X_{t_1}, X_{t_2}, \dots, X_{t_n}\}$ and $\{X_{t_1+k}, X_{t_2+k}, \dots, X_{t_n+k}\}$ are identical for all $t_1, t_2, \dots, t_n$ and $t_1 + k, t_2 + k, \dots, t_n + k$ in the time set J and all integers n . This means that the statistical properties of the process remain unchanged as time elapses.

For most cases of interest to us, it is enough for the time series to be weakly stationary. This is the case if the time series has a constant mean, constant variance and the covariance depends only on the lag.

We are also interested primarily in purely indeterministic processes. This means knowledge of the values $X_1, X_2, \dots, X_n$ is progressively less useful at predicting the value of X_N as $N \rightarrow \infty$.

We redefine the term 'stationary' to mean weakly stationary and purely indeterministic.

Importantly, the time series consisting of a sequence of white noise terms is weakly stationary and purely indeterministic. White noise is defined as a sequence of uncorrelated random variables. In time series, we assume that white noise has zero mean. This series has constant mean and variance, and covariance that depends only on whether the lag is zero or non-zero. It is purely indeterministic due to its random nature.

A time series process X is stationary if we can write it as a convergent sum of white noise terms.

It can be shown that this is equivalent to saying that the roots of the characteristic polynomial of the X terms are all greater than 1 in magnitude. For example, if the time series is defined by $X_t = \alpha_1 X_{t-1} + \dots + \alpha_p X_{t-p} + e_t + \beta_1 e_{t-1} + \dots + \beta_q e_{t-q}$ then the characteristic polynomial is $1 - \alpha_1 \lambda - \dots - \alpha_p \lambda^p$. To find the roots, we set this equal to zero and solve.

Invertibility

A time series process X is invertible if we can write the white noise term e_t as a convergent sum of the X terms.

It can be shown that this is equivalent to saying that the roots of the characteristic polynomial of the e terms are all greater than 1 in magnitude. For example, if the time series is given by $X_t = \alpha_1 X_{t-1} + \dots + \alpha_p X_{t-p} + e_t + \beta_1 e_{t-1} + \dots + \beta_q e_{t-q}$ then the characteristic polynomial of the e terms is $1 + \beta_1 \lambda + \dots + \beta_q \lambda^q$. To find the roots, we set this equal to zero and solve.

Invertibility is a desirable characteristic since it enables us to calculate the residual terms and hence analyse the goodness of fit of a particular model.

Backward shift and difference operators

The backwards shift operator, B , is defined as follows:

$$BX_t = X_{t-1}$$

$B\mu = \mu$ where μ is a constant.

The backward shift operator can be applied repeatedly so that $B^k X_t = X_{t-k}$.

The difference operator, ∇ , is defined as follows:

$$\nabla X_t = X_t - X_{t-1}$$

The difference operator can be applied repeatedly so that $\nabla^k X_t = \nabla^{k-1} X_t - \nabla^{k-1} X_{t-1}$.

The difference operator and backward shift operator are linked by $\nabla = 1 - B$.

Integrated processes

A time series process X is integrated of order d , denoted $I(d)$, if its d th difference is stationary. So X is $I(0)$ if the process X itself is stationary, and X is $I(1)$ if ∇X is stationary.

Autocovariance function

If the time series X is stationary, then its covariance depends only on the lag k . In this case, we define the autocovariance function as $\gamma_k = \text{cov}(X_t, X_{t+k})$.

Autocorrelation function

If the time series X is stationary, then its autocorrelation function (ACF) is:

$$\rho_k = \text{corr}(X_t, X_{t+k}) = \frac{\gamma_k}{\gamma_0}$$

For purely indeterministic time series processes X (where the past values of X become less useful the further into the future we look), $\rho_k \rightarrow 0$ as $k \rightarrow \infty$.

Partial autocorrelation function

The partial autocorrelation function (PACF), $\{\phi_k : k=1, 2, \dots\}$, is defined as the conditional correlation of X_{t+k} with X_t given $X_{t+1}, \dots, X_{t+k-1}$. Formulae for the partial autocorrelation are given on page 40 of the *Tables*.

Moving average processes

A time series process X is said to be $MA(q)$ (or moving average of order q) if it can be written as a weighted average of the past q white noise terms (plus a new white noise term):

$$X_t = e_t + \beta_1 e_{t-1} + \dots + \beta_q e_{t-q} \quad (\text{zero mean})$$

$$X_t = \mu + e_t + \beta_1 e_{t-1} + \dots + \beta_q e_{t-q} \quad (\text{mean } \mu)$$

Features of $MA(q)$ time series include:

- always stationary (as a finite sum of white noise)
- invertible if all the roots of the characteristic equation $1 + \beta_1 \lambda + \dots + \beta_q \lambda^q = 0$ are strictly greater than 1 in magnitude
- not Markov
- ρ_k cuts off for $k > q$
- ϕ_k decays geometrically as $k \rightarrow \infty$.

Autoregressive processes

A time series process X is said to be $AR(p)$ (or autoregressive of order p) if it depends on the past p terms of the series (plus a new white noise term):

$$X_t = \alpha_1 X_{t-1} + \dots + \alpha_p X_{t-p} + e_t \quad (\text{zero mean})$$

$$X_t = \mu + \alpha_1 (X_{t-1} - \mu) + \dots + \alpha_p (X_{t-p} - \mu) + e_t \quad (\text{mean } \mu)$$

Features of $AR(p)$ time series include:

- stationary if all the roots of the characteristic equation $1 - \alpha_1\lambda - \dots - \alpha_p\lambda^p = 0$ are strictly greater than 1 in magnitude
- always invertible
- only Markov if $p = 1$
- ρ_k decays geometrically as $k \rightarrow \infty$
- ϕ_k cuts off for $k > p$.

ARMA processes

A time series process X is said to be $ARMA(p, q)$ (or autoregressive moving average of order p, q) if it is the sum of an $AR(p)$ and an $MA(q)$ time series:

$$X_t = \alpha_1 X_{t-1} + \dots + \alpha_p X_{t-p} + e_t + \beta_1 e_{t-1} + \dots + \beta_q e_{t-q} \quad (\text{zero mean})$$

$$X_t = \mu + \alpha_1 (X_{t-1} - \mu) + \dots + \alpha_p (X_{t-p} - \mu) + e_t + \beta_1 e_{t-1} + \dots + \beta_q e_{t-q} \quad (\text{mean } \mu)$$

Features of $ARMA(p, q)$ time series include:

- stationary if all the roots of the characteristic equation of the AR part, ie $1 - \alpha_1\lambda - \dots - \alpha_p\lambda^p = 0$, are strictly greater than 1 in magnitude
- invertible if all the roots of the characteristic equation of the MA part, ie $1 + \beta_1\lambda + \dots + \beta_q\lambda^q = 0$, are strictly greater than 1 in magnitude
- only Markov if $p = 1, q = 0$
- ρ_k decays as $k \rightarrow \infty$
- ϕ_k decays as $k \rightarrow \infty$.

ARIMA processes

A time series process X is said to be $ARIMA(p, d, q)$ (or autoregressive integrated moving average of order p, d, q) if the d th difference $Y_t = \nabla^d X_t$ is a *stationary* $ARMA(p, q)$ time series process.



Chapter 13 Practice Questions

- 13.1 Give an expression for $2X_t - 5X_{t-1} + 4X_{t-2} - X_{t-3}$ in terms of second order differences.
- 13.2 A time series is defined by the relationship $X_t = X_{t-1} + Z_t$, where the Z_t are IID $N(0, \sigma^2)$ random variables.

Determine the relationship between $\text{var}(X_t)$ and $\text{var}(X_{t-1})$, and hence comment on the stationarity of this series.

- 13.3 Determine whether the process $X_n = X_{n-1} - \frac{1}{2}X_{n-2} + e_n$ is stationary.
- 13.4 An autoregressive stationary time series W_t is defined by the relationship:

$$W_t = 0.6W_{t-1} + 0.4W_{t-2} - 0.1W_{t-3} + Z_t$$

for integer times t , where $\{Z_t\}$ represents a set of uncorrelated random variables with mean 0 and variance σ^2 .

- (i) Explain why $\text{cov}(W_t, Z_{t+1}) = 0$ and $\text{cov}(W_{t-1}, W_t) = \text{cov}(W_{t-1}, W_{t-2})$.
- (ii) By considering $\text{cov}(W_t, W_{t-k})$ when $k = 0, 1, 2, 3$, write down a set of four equations relating the values of the autocovariance function γ_k at lags $k = 0, 1, 2, 3$.
- (iii) Solve the four equations in part (ii) to find both the autocovariance function and the autocorrelation function for lags 0, 1, 2 and 3.
- 13.5 Calculate the autocorrelation function of the process $X_n = 1 + e_n - 5e_{n-1} + 6e_{n-2}$.
- 13.6 (i) The first differences of a time series X can be modelled by the process:

$$\nabla X_n = 0.5\nabla X_{n-1} + e_n$$

Determine the model for X_n .

- (ii) Show that the process X_n is non-stationary.
- 13.7 (i) Show that the relationship $Y_t = 0.7Y_{t-1} + 0.3Y_{t-2} + Z_t + 0.7Z_{t-1}$ (where the Z 's denote white noise) defines an $ARIMA(1,1,1)$ process.
- (ii) Show carefully that the relationship $S_t = 1.5S_{t-1} + 0.5S_{t-3} + Z_t + 0.5Z_{t-1}$ cannot be expressed as an $ARIMA(1,2,1)$ process.
- 13.8 Consider the process with defining equation:

$$X_n = 5X_{n-1} - 4X_{n-2} + X_{n-3} + e_n$$

Write this as a vector process that possesses the Markov property.

13.9 Let $X_n = e_n + e_{n-2}$ be an MA(2) process where $e_n \sim N(0,1)$.

- (i) Calculate $P(X_n \geq 0 | X_{n-1} \leq 0)$.
- (ii) Compare your answer to (i) with $P(X_n \geq 0 | X_{n-1} \leq 0, X_{n-2} \leq 0)$ and hence comment on whether the process is Markov.

13.10 Calculate the values of ρ_1 and ρ_2 , the autocorrelation function at lags 1 and 2, for the stationary AR(2) process defined by the equation:

Exam style

$$X_n = -0.8X_{n-1} + 0.1X_{n-2} + \varepsilon_n \quad [4]$$

13.11 Consider the time series model defined by:

Exam style

$$X_t = \alpha_1 X_{t-1} + \alpha_2 X_{t-2} + \alpha_3 X_{t-3} + \varepsilon_t$$

where $\{\varepsilon_t\}$ is white noise.

- (i) Show that the autocorrelation coefficient with lag 1 for this process is:

$$\rho_1 = \frac{\alpha_1 + \alpha_2 \alpha_3}{1 - \alpha_2 - \alpha_1 \alpha_3 - \alpha_3^2} \quad [3]$$

- (ii) Consider the case where $\alpha_1 = \alpha_2 = \alpha_3 = 0.2$.

- (a) Comment on the stationarity of this model.

$$\text{Hint: } 5 - x - x^2 - x^3 \approx (1.278 - x)(3.912 + 2.278x + x^2)$$

- (b) Calculate ρ_1 and ρ_2 .
- (c) Calculate the partial autocorrelation coefficients ϕ_1 and ϕ_2 .
- (d) Sketch correlograms of the autocorrelation function and the partial autocorrelation function. (You are not required to calculate the coefficients for higher lags.)

[9]

[Total 12]

13.12 $\{X_t\}$ is a stationary $ARMA(1,2)$ time series defined at integer times by the relationship:

Exam style

$$X_t = \alpha X_{t-1} + Z_t + \beta Z_{t-2}$$

where α, β are constants and $\{Z_t\}$ is a purely random process with mean 0 and constant variance σ^2 .

(i) Define the term 'weakly stationary process'. [2]

(ii) Assuming that the above process has a very long history, state the conditions on α and β needed to ensure that it is:

(a) stationary

(b) invertible. [2]

(iii) Show that for any integer s :

$$\text{cov}(X_s, Z_s) = \sigma^2 \quad \text{cov}(X_s, Z_{s-1}) = \alpha\sigma^2 \quad \text{cov}(X_s, Z_{s-2}) = (\alpha^2 + \beta)\sigma^2 \quad [3]$$

(iv) (a) By considering $\text{cov}(X_t, X_t)$, $\text{cov}(X_t, X_{t-1})$ and $\text{cov}(X_t, X_{t-2})$, write down three equations involving γ_0 , γ_1 and γ_2 .

(b) Hence find expressions for γ_0 , γ_1 and γ_2 in terms of the parameters α , β and σ^2 . [7]

(v) (a) Calculate the values of ρ_0 , ρ_1 , ρ_2 and ρ_3 in the case where $\alpha = -0.4$ and $\beta = -0.9$.

(b) Hence sketch a graph of the autocorrelation function ρ_k for lags $k = 0, 1, 2, \dots, 10$ in this case. [5]

[Total 19]

The solutions start on the next page so that you can
separate the questions and solutions.



Chapter 13 Solutions

13.1 Using the difference operator, we can write:

$$\begin{aligned}
 2X_t - 5X_{t-1} + 4X_{t-2} - X_{t-3} &= 2(X_t - X_{t-1}) - 3(X_{t-1} - X_{t-2}) + (X_{t-2} - X_{t-3}) \\
 &= 2\nabla X_t - 3\nabla X_{t-1} + \nabla X_{t-2} \\
 &= 2(\nabla X_t - \nabla X_{t-1}) - (\nabla X_{t-1} - \nabla X_{t-2}) \\
 &= 2\nabla^2 X_t - \nabla^2 X_{t-1}
 \end{aligned}$$

13.2 The series is defined by the equation:

$$X_t = X_{t-1} + Z_t$$

If we take variances of both sides, we get:

$$\text{var}(X_t) = \text{var}(X_{t-1}) + \sigma^2$$

Here we have used the fact that Z_t and X_{t-1} are uncorrelated. This is because X_{t-1} depends only on the past values of X and Z (ie values for times up to $t-1$), whereas Z_t is a new random value at time t that is not influenced in any way by these earlier values.

This implies that $\text{var}(X_t) > \text{var}(X_{t-1})$, which contradicts the stationarity requirement, which requires (amongst other things) that the variance is independent of time. So the series cannot be stationary.

13.3 The characteristic equation of the AR part is:

$$1 - \lambda + 0.5\lambda^2 = 0$$

The roots of this equation are:

$$\lambda = \frac{-(-1) \pm \sqrt{(-1)^2 - 4 \times 0.5 \times 1}}{2 \times 0.5} = \frac{1 \pm \sqrt{-1}}{1} = 1 \pm i$$

Since $|1 \pm i| = \sqrt{2} > 1$, the process is stationary.

13.4 (i) W_t is only dependent on past Z 's, which are not correlated with the future Z 's. Therefore $\text{cov}(W_t, Z_{t+1}) = 0$. Also, since W_t is stationary, only the lag matters so the second equation holds.

- (ii) If we take the covariance of both sides of the defining equation with W_t, W_{t-1}, W_{t-2} and W_{t-3} in turn, then we obtain the four equations:

$$\gamma_0 = 0.6\gamma_1 + 0.4\gamma_2 - 0.1\gamma_3 + \sigma^2 \quad (1)$$

$$\gamma_1 = 0.6\gamma_0 + 0.4\gamma_1 - 0.1\gamma_2 \quad (2)$$

$$\gamma_2 = 0.6\gamma_1 + 0.4\gamma_0 - 0.1\gamma_1 \quad (3)$$

$$\gamma_3 = 0.6\gamma_2 + 0.4\gamma_1 - 0.1\gamma_0 \quad (4)$$

The σ^2 term in the first equation comes from:

$$\text{cov}(W_t, Z_t) = \text{cov}(0.6W_{t-1} + 0.4W_{t-2} - 0.1W_{t-3} + Z_t, Z_t) = \text{cov}(Z_t, Z_t) = \sigma^2$$

- (iii) Equation (2) gives $\gamma_2 = 6\gamma_0 - 6\gamma_1$.

Likewise, from Equation (3) we get $\gamma_2 = 0.4\gamma_0 + 0.5\gamma_1$.

Equating these two expressions for γ_2 gives $\gamma_1 = \frac{56}{65}\gamma_0$ and hence $\gamma_2 = \frac{54}{65}\gamma_0$.

Equation (4) can then be applied to get $\gamma_3 = \frac{483}{650}\gamma_0$.

Finally Equation (1) can be applied to show that $\sigma^2 = 0.22508\gamma_0$.

Rearranging these expressions in terms of σ^2 we obtain the first four autocovariance values:

$$\gamma_0 = 4.4429\sigma^2, \quad \gamma_1 = 3.8278\sigma^2, \quad \gamma_2 = 3.6910\sigma^2, \quad \gamma_3 = 3.3014\sigma^2$$

The autocorrelations are then obtained by dividing by γ_0 :

$$\rho_0 = 1, \quad \rho_1 = 0.862, \quad \rho_2 = 0.831, \quad \rho_3 = 0.743$$

13.5 The autocovariance function is:

$$\text{cov}(X_n, X_{n+k}) = \text{cov}(e_n - 5e_{n-1} + 6e_{n-2}, e_{n+k} - 5e_{n+k-1} + 6e_{n+k-2})$$

$$= \begin{cases} 62 & k=0 \\ -35 & |k|=1 \\ 6 & |k|=2 \\ 0 & |k|>2 \end{cases}$$

For example:

$$\begin{aligned}
 \gamma_0 &= \text{cov}(X_n, X_n) \\
 &= \text{cov}(e_n - 5e_{n-1} + 6e_{n-2}, e_n - 5e_{n-1} + 6e_{n-2}) \\
 &= \text{cov}(e_n, e_n) + \text{cov}(-5e_{n-1}, -5e_{n-1}) + \text{cov}(6e_{n-2}, 6e_{n-2}) \\
 &= 1^2 + (-5)^2 + 6^2 = 62
 \end{aligned}$$

and similarly for the other values of k .

Since the autocovariance and autocorrelation functions are even, we can give the results for non-negative values of k . The autocorrelation function is:

$$\rho_0 = 1 \quad \rho_{\pm 1} = -\frac{35}{62} \quad \rho_{\pm 2} = \frac{6}{62} \quad \rho_k = 0 \text{ for } |k| > 2$$

13.6 (i) **Integrated process**

The series $\nabla X_n = 0.5\nabla X_{n-1} + e_n$, can be written:

$$X_n - X_{n-1} = 0.5(X_{n-1} - X_{n-2}) + e_n$$

Rearranging we end up with:

$$X_n = 1.5X_{n-1} - 0.5X_{n-2} + e_n$$

(ii) **Show that X is non-stationary**

We have:

$$X_n - 1.5X_{n-1} + 0.5X_{n-2} = e_n$$

which is an $AR(2)$ process. Its characteristic equation is:

$$1 - 1.5\lambda + 0.5\lambda^2 = 0$$

This has roots $\frac{1.5 \pm \sqrt{2.25 - 2}}{1}$, ie 1 and 2. For stationarity, we require both roots to be strictly greater than 1 in magnitude. This is not the case here, so the process is not stationary.

13.7 (i) **ARIMA(1,1,1)**

The characteristic equation of the AR part is:

$$1 - 0.7\lambda - 0.3\lambda^2 = 0$$

The roots of this equation are 1 and $-\frac{10}{3}$. Both roots must be strictly greater than 1 in magnitude for the series to be stationary. This is not the case here.

Differencing once removes the root of 1 as we can see by subtracting Y_{t-1} from both sides of the defining equation. This gives:

$$\begin{aligned} Y_t - Y_{t-1} &= -0.3Y_{t-1} + 0.3Y_{t-2} + Z_t + 0.7Z_{t-1} \\ &= -0.3(Y_{t-1} - Y_{t-2}) + Z_t + 0.7Z_{t-1} \end{aligned}$$

This can be expressed using the difference operator as:

$$\nabla Y_t = -0.3\nabla Y_{t-1} + Z_t + 0.7Z_{t-1}$$

The characteristic equation of the AR part of the differenced series is:

$$1 + 0.3\lambda = 0$$

and the root of this equation is $-\frac{10}{3}$. So ∇Y is stationary $ARMA(1,1)$ and hence Y is $ARIMA(1,1,1)$.

(ii) **Not an $ARIMA(1,2,1)$**

The general form of an $ARIMA(1,2,1)$ model is:

$$\nabla^2 S_t = \alpha \nabla^2 S_{t-1} + Z_t + \beta Z_{t-1}$$

Expanding the ∇ 's gives:

$$S_t - 2S_{t-1} + S_{t-2} = \alpha(S_{t-1} - 2S_{t-2} + S_{t-3}) + Z_t + \beta Z_{t-1}$$

$$\text{ie } S_t = (2 + \alpha)S_{t-1} - (1 + 2\alpha)S_{t-2} + \alpha S_{t-3} + Z_t + \beta Z_{t-1}$$

The relationship given is:

$$S_t = 1.5S_{t-1} + 0.5S_{t-3} + Z_t + 0.5Z_{t-1}$$

Comparing the coefficients of S_{t-1} implies that $\alpha = -0.5$. This correctly makes the S_{t-2} term vanish, but gives the wrong sign for S_{t-3} . So this definition is not consistent with an $ARIMA(1,2,1)$ model.

13.8 We can write $X_n = 5X_{n-1} - 4X_{n-2} + X_{n-3} + e_n$ in vector form as follows:

$$\begin{pmatrix} X_n \\ X_{n-1} \\ X_{n-2} \end{pmatrix} = \begin{pmatrix} 5 & -4 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} X_{n-1} \\ X_{n-2} \\ X_{n-3} \end{pmatrix} + \begin{pmatrix} e_n \\ 0 \\ 0 \end{pmatrix}$$

This vector-valued process has the Markov property.

- 13.9 (i) If e_n and e_{n-2} are independent standard normal random variables, then $e_n + e_{n-2} \sim N(0, 2)$. Hence:

$$\begin{aligned} P(X_n \geq 0 | X_{n-1} \leq 0) &= P(e_n + e_{n-2} \geq 0 | e_{n-1} + e_{n-3} \leq 0) \\ &= P(e_n + e_{n-2} \geq 0) \\ &= 0.5 \end{aligned}$$

- (ii) We have:

$$\begin{aligned} P(X_n \geq 0 | X_{n-1} \leq 0, X_{n-2} \leq 0) &= P(e_n + e_{n-2} \geq 0 | e_{n-1} + e_{n-3} \leq 0, e_{n-2} + e_{n-4} \leq 0) \\ &= P(e_n + e_{n-2} \geq 0 | e_{n-2} + e_{n-4} \leq 0) \\ &= \frac{P(e_n + e_{n-2} \geq 0 \cap e_{n-2} + e_{n-4} \leq 0)}{P(e_{n-2} + e_{n-4} \leq 0)} \\ &= \frac{P(e_n + e_{n-2} \geq 0 \cap e_{n-2} + e_{n-4} \leq 0)}{0.5} \end{aligned}$$

The process is Markov if $P(X_n \geq 0 | X_{n-1} \leq 0) = P(X_n \geq 0 | X_{n-1} \leq 0, X_{n-2} \leq 0)$. Hence, we require $P(e_n + e_{n-2} \geq 0 \cap e_{n-2} + e_{n-4} \leq 0) = 0.25$.

Equivalently, we require $P(e_n \geq -e_{n-2} \cap e_{n-4} \leq -e_{n-2}) = 0.25$.

For any given observation ε of e_{n-2} , this is:

$$(1 - \Phi(-\varepsilon))\Phi(-\varepsilon) = 0.25$$

and the solution to this quadratic is at $\Phi(-\varepsilon) = 0.5$ or $e_{n-2} = 0$.

In other words, $P(X_n \geq 0 | X_{n-1} \leq 0) = P(X_n \geq 0 | X_{n-1} \leq 0, X_{n-2} \leq 0)$ only if $e_{n-2} = 0$. So the process cannot be Markov.

- 13.10 The defining equation is:

$$X_n = -0.8X_{n-1} + 0.1X_{n-2} + \varepsilon_n$$

The autocovariance at lag 1 is:

$$\begin{aligned} \gamma_1 &= \text{cov}(X_n, X_{n-1}) \\ &= \text{cov}(-0.8X_{n-1} + 0.1X_{n-2} + \varepsilon_n, X_{n-1}) \\ &= -0.8\text{cov}(X_{n-1}, X_{n-1}) + 0.1\text{cov}(X_{n-2}, X_{n-1}) + \text{cov}(\varepsilon_n, X_{n-1}) \\ &= -0.8\gamma_0 + 0.1\gamma_1 \end{aligned}$$

[1]

Dividing through by γ_0 and using the fact that $\rho_0 = 1$ gives:

$$\rho_1 = -0.8 + 0.1\rho_1 \Rightarrow \rho_1 = -\frac{8}{9} \quad [1]$$

Similarly, the autocovariance at lag 2 is:

$$\begin{aligned} \gamma_2 &= \text{cov}(X_n, X_{n-2}) \\ &= \text{cov}(-0.8X_{n-1} + 0.1X_{n-2} + \varepsilon_n, X_{n-2}) \\ &= -0.8\gamma_1 + 0.1\gamma_0 \end{aligned} \quad [1]$$

Dividing through by γ_0 gives:

$$\rho_2 = -0.8\rho_1 + 0.1 = -0.8 \times \left(-\frac{8}{9}\right) + 0.1 = \frac{73}{90} \quad [1]$$

The value of γ_0 is not required here.

13.11 (i) **Autocorrelation coefficient**

Applying the Yule-Walker method by taking covariances with X_{t-1} , we get:

$$\text{cov}(X_t, X_{t-1}) = \text{cov}(\alpha_1 X_{t-1} + \alpha_2 X_{t-2} + \alpha_3 X_{t-3} + \varepsilon_t, X_{t-1})$$

$$\text{ie: } \gamma_1 = \alpha_1\gamma_0 + \alpha_2\gamma_1 + \alpha_3\gamma_2 + 0 \quad [\frac{1}{2}]$$

Dividing by γ_0 :

$$\rho_1 = \alpha_1 + \alpha_2\rho_1 + \alpha_3\rho_2$$

$$\text{ie: } (1 - \alpha_2)\rho_1 = \alpha_1 + \alpha_3\rho_2 \quad [\frac{1}{2}]$$

Then taking covariances with X_{t-2} :

$$\text{cov}(X_t, X_{t-2}) = \text{cov}(\alpha_1 X_{t-1} + \alpha_2 X_{t-2} + \alpha_3 X_{t-3} + \varepsilon_t, X_{t-2})$$

$$\text{ie: } \gamma_2 = \alpha_1\gamma_1 + \alpha_2\gamma_0 + \alpha_3\gamma_1 + 0 \quad [\frac{1}{2}]$$

Dividing by γ_0 :

$$\rho_2 = \alpha_1\rho_1 + \alpha_2 + \alpha_3\rho_1$$

$$\text{ie: } \rho_2 = (\alpha_1 + \alpha_3)\rho_1 + \alpha_2 \quad [\frac{1}{2}]$$

If we eliminate ρ_2 from these two simultaneous equations, we get:

$$(1 - \alpha_2)\rho_1 = \alpha_1 + \alpha_3[(\alpha_1 + \alpha_3)\rho_1 + \alpha_2] \quad [\frac{1}{2}]$$

Rearranging:

$$[(1 - \alpha_2) - \alpha_3(\alpha_1 + \alpha_3)]\rho_1 = \alpha_1 + \alpha_3\alpha_2$$

So:

$$\rho_1 = \frac{\alpha_1 + \alpha_3 \alpha_2}{(1 - \alpha_2) - \alpha_3(\alpha_1 + \alpha_3)} = \frac{\alpha_1 + \alpha_2 \alpha_3}{1 - \alpha_2 - \alpha_1 \alpha_3 - \alpha_3^2} \quad [\frac{1}{2}]$$

(ii)(a) **Stationarity**

In this case the model can be written as:

$$X_t - 0.2X_{t-1} - 0.2X_{t-2} - 0.2X_{t-3} = \varepsilon_t$$

Stationarity is determined by the nature of the roots of the characteristic equation:

$$1 - 0.2\lambda - 0.2\lambda^2 - 0.2\lambda^3 = 0 \quad [1]$$

Multiplying by 5, this is:

$$5 - \lambda - \lambda^2 - \lambda^3 = 0$$

Using the hint given, we can write this as:

$$(1.278 - \lambda)(3.912 + 2.278\lambda + \lambda^2) = 0$$

$$\Rightarrow \lambda = 1.278 \quad \text{or} \quad 3.912 + 2.278\lambda + \lambda^2 = 0 \quad [\frac{1}{2}]$$

The quadratic equation has roots:

$$\lambda = \frac{-2.278 \pm \sqrt{(2.278)^2 - 4(1)(3.912)}}{2} = -1.139 \pm 1.617i \quad [1]$$

The real root (1.278) and the two complex roots are all strictly greater than 1 in magnitude.

(Recall that the magnitude of a complex root $a + bi$ is given by $\sqrt{a^2 + b^2}$.)

So this model is stationary. [1]

(ii)(b) **ACF at lags 1 and 2**

Using the formula derived in part (i):

$$\rho_1 = \frac{\alpha_1 + \alpha_2 \alpha_3}{1 - \alpha_2 - \alpha_1 \alpha_3 - \alpha_3^2} = \frac{0.2 + (0.2)^2}{1 - 0.2 - (0.2)^2 - (0.2)^2} = \frac{0.24}{0.72} = \frac{1}{3} \quad [\frac{1}{2}]$$

From the second Yule-Walker equation, derived in part (i), we also have:

$$\rho_2 = (\alpha_1 + \alpha_3)\rho_1 + \alpha_2 = (0.2 + 0.2) \times \frac{1}{3} + 0.2 = \frac{1}{3} \quad [\frac{1}{2}]$$

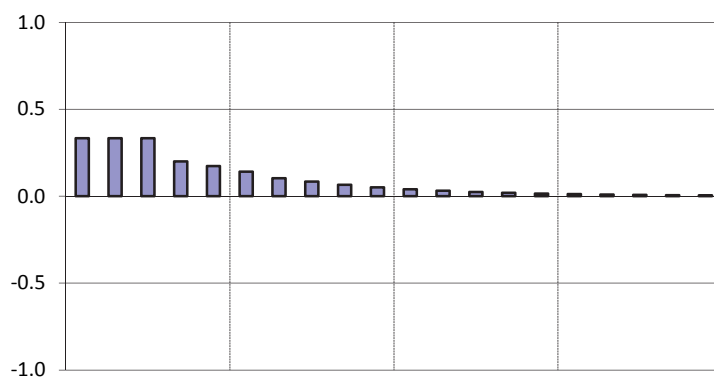
(ii)(c) **PACFs at lags 1 and 2**

Using the formulae on page 40 of the *Tables*:

$$\phi_1 = \rho_1 = \frac{1}{3} \quad \text{and} \quad \phi_2 = \frac{\rho_2 - \rho_1^2}{1 - \rho_1^2} = \frac{1}{4} \quad [1]$$

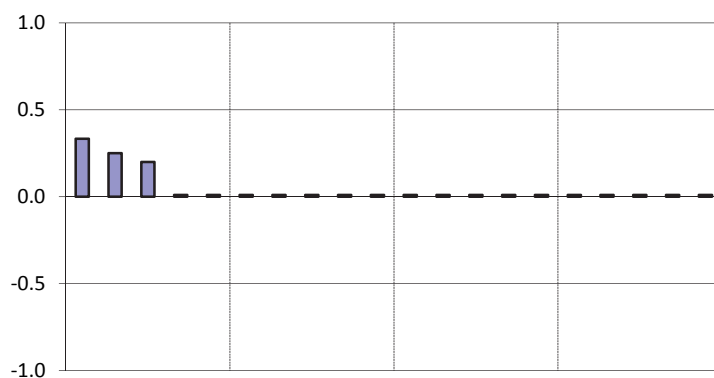
(ii)(d) **Correlograms**

We've already calculated $\rho_1 = \frac{1}{3}$ and $\rho_2 = \frac{1}{3}$. Higher-lag autocorrelations will satisfy the Yule-Walker equation $\rho_k = 0.2(\rho_{k-1} + \rho_{k-2} + \rho_{k-3})$. So the values will tail off quite quickly to zero, always taking positive values. The correlogram (*ie* graph of ρ_k versus k) will look like this:



[2]

We've already calculated $\phi_1 = \frac{1}{3}$ and $\phi_2 = \frac{1}{4}$. Since the process is $AR(3)$, the partial autocorrelations will all equal zero from lag 4 onwards. So the partial correlogram (*ie* graph of ϕ_k versus k) will look like this:



[2]

13.12 (i) **Weakly stationary process**

A weakly stationary process has constant mean, and the covariance is constant for each fixed lag. The variance (a special case of the covariance) is also constant. [2]

(ii)(a) **Condition for stationarity**

The root of the characteristic equation of the AR part, ie:

$$1 - \alpha\lambda = 0$$

must be strictly greater than 1 in magnitude. So, for stationarity, we must have $|\alpha| < 1$.

(ii)(b) **Condition for invertibility**

The root of the characteristic equation of the MA part, ie:

$$1 - \beta\lambda^2 = 0$$

must be strictly greater than 1 in magnitude. So, for invertibility, we must have $|\beta| < 1$. [2]

(iii) **Covariances**

Using the definition of the series with $t = s$, we have:

$$\begin{aligned} \text{cov}(X_s, Z_s) &= \text{cov}(\alpha X_{s-1} + Z_s + \beta Z_{s-2}, Z_s) \\ &= \alpha \text{cov}(X_{s-1}, Z_s) + \text{cov}(Z_s, Z_s) + \beta \text{cov}(Z_{s-2}, Z_s) \end{aligned} \quad [1/2]$$

The first term on the RHS is zero since Z_s is uncorrelated with earlier values of the series. The third term is also zero since $\{Z_t\}$ is a purely random process. So:

$$\text{cov}(X_s, Z_s) = \text{cov}(Z_s, Z_s) = \text{var}(Z_s) = \sigma^2 \quad [1/2]$$

Similarly using the result just proved, with s replaced by $s - 1$, we have:

$$\begin{aligned} \text{cov}(X_s, Z_{s-1}) &= \text{cov}(\alpha X_{s-1} + Z_s + \beta Z_{s-2}, Z_{s-1}) \\ &= \alpha \text{cov}(X_{s-1}, Z_{s-1}) = \alpha \sigma^2 \end{aligned} \quad [1]$$

Finally:

$$\begin{aligned} \text{cov}(X_s, Z_{s-2}) &= \text{cov}(\alpha X_{s-1} + Z_s + \beta Z_{s-2}, Z_{s-2}) \\ &= \alpha \text{cov}(X_{s-1}, Z_{s-2}) + \text{cov}(Z_s, Z_{s-2}) + \beta \text{cov}(Z_{s-2}, Z_{s-2}) \\ &= \alpha \text{cov}(X_s, Z_{s-1}) + 0 + \beta \sigma^2 \\ &= (\alpha^2 + \beta) \sigma^2 \end{aligned} \quad [1]$$

(iv)(a) **Yule-Walker equations**

If we replace one of the X_t 's in $\text{cov}(X_t, X_t)$ with the definition given for X_t , we get:

$$\text{cov}(X_t, X_t) = \text{cov}(\alpha X_{t-1} + Z_t + \beta Z_{t-2}, X_t)$$

Using the first and third results from part (iii) gives:

$$\gamma_0 = \alpha\gamma_1 + \sigma^2 + \beta(\alpha^2 + \beta)\sigma^2 = \alpha\gamma_1 + (1 + \alpha^2\beta + \beta^2)\sigma^2 \quad \dots (1) \quad [1]$$

If we replace the X_t in $\text{cov}(X_t, X_{t-1})$ with the definition given for X_t , we get:

$$\text{cov}(X_t, X_{t-1}) = \text{cov}(\alpha X_{t-1} + Z_t + \beta Z_{t-2}, X_{t-1}) \quad [1]$$

Using the second result from part (iii), this simplifies to:

$$\gamma_1 = \alpha\gamma_0 + 0 + \alpha\beta\sigma^2 = \alpha\gamma_0 + \alpha\beta\sigma^2 \quad \dots (2) \quad [1]$$

Similarly:

$$\text{cov}(X_t, X_{t-2}) = \text{cov}(\alpha X_{t-1} + Z_t + \beta Z_{t-2}, X_{t-2})$$

Using the first result from part (iii), this simplifies to:

$$\gamma_2 = \alpha\gamma_1 + 0 + \beta\sigma^2 = \alpha\gamma_1 + \beta\sigma^2 \quad \dots (3) \quad [1]$$

(iv)(b) **Obtain autocovariances**

Substituting the value of γ_1 from Equation (2) into Equation (1), rearranging and simplifying gives:

$$\gamma_0 = \frac{1 + \beta^2 + 2\alpha^2\beta}{1 - \alpha^2} \sigma^2 \quad [1]$$

Substituting this into Equation (2) gives:

$$\gamma_1 = \frac{\alpha(1 + \beta + \beta^2 + \alpha^2\beta)}{1 - \alpha^2} \sigma^2 \quad [1]$$

Substituting this into Equation (3) gives:

$$\gamma_2 = \frac{\beta + \alpha^2 + \alpha^2\beta^2 + \alpha^4\beta}{1 - \alpha^2} \sigma^2 \quad [1]$$

(v)(a) **ACF**

Using the values given for α and β , the numerators in the expressions for γ_0 , γ_1 and γ_2 are 1.522, -0.3064 and -0.63344.

So:

$$\rho_0 = 1 \quad \rho_1 = -\frac{0.3064}{1.522} = -0.201 \quad \rho_2 = -\frac{0.63344}{1.522} = -0.416 \quad [2]$$

If we use the same method as in (iv)(a), we see that subsequent values of γ (and hence ρ) will just include an extra factor of α each time.

So:

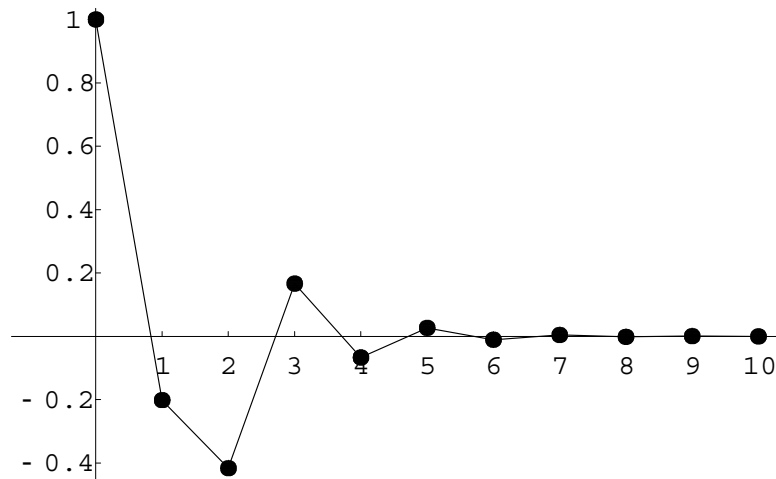
$$\rho_k = -0.416 \times (-0.4)^{k-2} \text{ for } k \geq 2$$

$$\Rightarrow \rho_3 = -0.416 \times (-0.4) = 0.166$$

[1]

(v)(b) **Graph of ACF**

So the graph of ρ_k looks like this (with the values for $k \geq 2$ alternating in sign and reducing in magnitude):



[2]

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